

Firms and the Evolution of Earnings Inequality in Argentina's Volatile Economy*

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Abstract

Earnings inequality across Latin America fell sharply during the 2000s, then stagnated. We ask how much of this reversal operated through firms. Using matched employer-employee records for Argentina's formal sector over 1995–2019, spanning crisis, boom, and stagnation, we estimate two-way fixed effects models at five-year intervals and decompose the variance of log earnings. Firm wage effects account for about a third of the 2000s decline and more than account for the 2010s increase, while sorting plays a minor role throughout. The rise in firm-effect dispersion is concentrated among small firms, and 45.2% of it reflects widening gaps between industries, which track sectoral productivity. Negotiated wage floors rose sharply in the 2000s, compressing firm effects from below, but lost force thereafter, and the association between informality and firm-effect dispersion strengthened in the 2010s. The evidence suggests that the 2010s stagnation widened inequality primarily by allowing low-paying firms to fall behind.

Keywords: earnings inequality, firm wage effects, AKM, collective bargaining, informality, Argentina.

JEL codes: E24, J31, J38, J46, J51, O15.

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1 Introduction

Latin America experienced a substantial decline in earnings inequality in the 2000s (e.g., [Gasparini and Lustig 2011](#)), followed by a slowdown and, in some cases, a reversal in the 2010s (e.g., [Gasparini et al. 2016](#); [Messina and Silva 2019](#); [Alvaredo et al. 2025](#)). Research on the decline has emphasized changes in educational attainment and the returns to education (e.g., [Manacorda et al. 2010](#); [Lustig et al. 2016](#); [Fernández and Messina 2018](#)), alongside a compression of pay differences across firms (e.g., [Alvarez et al. 2018](#)) and stronger labor market institutions (e.g., [Maurizio and Vazquez 2016](#); [Engbom and Moser 2022](#)). Less is known about the role of firms in the subsequent stagnation and reversal of inequality reduction. Clarifying this role would help assess the scope of worker- and firm-oriented policies ([OECD 2021](#)) and their effectiveness across macroeconomic regimes.

Firm wage premia may be particularly sensitive to macroeconomic conditions in developing economies. Productivity dispersion across firms is large in these settings, reflecting in part distortions in the allocation of resources ([Hsieh and Klenow 2009](#); [Busso et al. 2010](#)). Because more productive firms tend to pay higher wage premia, this heterogeneity can translate into larger wage dispersion across firms (e.g., [Card et al. 2018](#); [Alvarez et al. 2018](#); [Bassier 2023](#)). Labor market institutions, such as higher minimum wages and stronger collective bargaining, tend to compress dispersion in firm wage premia (e.g., [Card et al. 2013](#); [Bergolo et al. 2025](#)). Against this background, transitions from growth to stagnation may affect firms unevenly and disrupt resource allocation across sectors ([Cavallo et al. 2010](#); [Oberfield 2013](#)). One possibility is that less productive firms have less capacity to absorb adverse conditions, causing their wage premia to fall. Weak enforcement may further amplify these effects, as firms respond to adverse conditions by evading labor regulations, potentially widening the dispersion of firm wage effects ([Ulyssea 2018](#); [Ponczek and Ulyssea 2021](#)).

Quantifying the contribution of firms to earnings inequality and how it relates to macroeconomic and institutional changes is challenging for three main reasons. First, it requires administrative data tracking worker mobility across firms. Second, the data must span a long period in an economy undergoing significant macroeconomic fluctuations. Third, investigating the mechanisms requires additional data on firm characteristics, sectoral productivity, and labor market institutions, which are rarely available in developing countries.

This paper studies the role of firms in the evolution of earnings inequality in Argentina between 1995 and 2019, a case study that allows us to address these challenges, revealing new insights into the reversal of inequality reduction in the 2010s. First, a rich administrative dataset covering registered employment allows us to identify the role of firms. Second, the 25-year sample spans crises, expansion, and stagnation. Third, newly digitized data on wage

floors negotiated in collective bargaining, household survey data, and newly compiled data on sectoral productivity allow us to explore the drivers of dispersion of firm wage premia. Our main finding is that firm wage effects more than account for the 2010s increase in earnings inequality. The rise in firm-premium dispersion is concentrated among smaller firms, with nearly half reflecting widening gaps between industries. We relate these patterns to sectoral productivity, wage-setting institutions, and informality across local labor markets.

We first present descriptive evidence on the evolution of earnings inequality and the role of firms in these trends. Earnings inequality rose during the 1990s, fell sharply during the 2000s, and increased again during the 2010s. While the 2000s decline was broad-based, the subsequent increase was concentrated at the bottom of the earnings distribution. A descriptive variance decomposition shows that differences between firms account for approximately two-thirds of the variance of log earnings and drive its fluctuations over time. Around half of the between-firm variance reflects industry differences, while most of its increase during the 2010s occurred within, rather than between, firm-size groups.

To distinguish firm wage premia from differences in workforce composition, we estimate the two-way fixed effects model of [Abowd et al. \(1999\)](#) (AKM hereafter) in five-year intervals. The model isolates the contribution of workers, firms, and assortative matching between them to the variance of log earnings. We use worker mobility to identify firm wage effects and present diagnostic evidence supporting the model’s identifying assumptions, following [Card et al. \(2013\)](#). Our baseline variance estimates correct for limited mobility bias ([Kline et al. 2020](#)), and the main time patterns are similar across corrected and uncorrected estimates.

The AKM decomposition shows sharp differences in the sources of inequality fluctuations in the 2000s and 2010s. Firm effects account for 31% of the decline, while worker effects account for 60%. In contrast, during the 2010s the variance of firm effects increased more than twice as much as the overall earnings variance, more than accounting for the reversal of inequality reduction. The increase in firm wage dispersion is most pronounced among small firms and at the bottom of the firm-premium distribution, and is widespread across regions. By 2015–19, the firm component accounted for 41.5% of the variance of log earnings, compared with 33.0% in 1995–99, and exceeded the worker component. As we discuss below, this share is large in international perspective. By contrast, changes in assortative matching (or sorting) played only a minor role in the evolution of earnings inequality.

We then examine how the transition from expansion to stagnation changed the conditions under which firms set wages. Labor productivity rose during the 2000s expansion, but productivity declined in the 2010s and productivity gaps across sectors widened. Constructing industry wage differentials from estimated firm effects following [Card et al. \(2024\)](#), we find that widening gaps between industries account for nearly half of the increase in firm-effect

variance during the 2010s. Less productive sectors pay lower firm wage premia on average and exhibit greater dispersion across firms within the sector. This negative association between productivity and within-sector dispersion is stronger in 2015–19 than in 2005–09. These patterns suggest that uneven productivity performance may have contributed to widening firm pay differences both between and within sectors.¹

Rising wage floors compressed within-sector firm pay differences during the 2000s expansion, but this compression weakened during the stagnation. Using newly digitized data on negotiated wage floors, we instrument wage-floor growth with industries’ differential exposure to the increase in the national minimum wage in 2004–07 and find that higher floors reduced the dispersion of firm wage effects between 1995–99 and 2005–09. Wage floors raised the lower percentiles of the firm-effect distribution, with little effect at the top. The estimated effect of the 2000s wage-floor increase on dispersion is smaller and statistically indistinguishable from zero by 2015–19. Over this period, the real minimum wage eroded, while negotiated floors eventually stopped rising. Weak compliance may have further limited their ability to protect pay at the bottom as economic conditions deteriorated.

Consistent with this possibility, local labor markets with higher informality rates experienced larger increases in the variance of firm wage effects, with the divergence most pronounced during the 2010s. The widening came from lower percentiles falling behind, with little differential change at the top. These markets also experienced a stronger increase in overall earnings inequality and a relative shift toward smaller formal firms. These patterns are consistent with firms responding to adverse macroeconomic conditions by reducing firm size and labor costs, with weak enforcement providing scope for such adjustments.²

Our findings suggest a hypothesis for the reversal of inequality reduction in Latin America: the gains supported by economic expansion may prove fragile when macroeconomic conditions deteriorate. In Argentina, widening productivity gaps and weakening labor market institutions coincided with low-paying firms falling behind, particularly in local labor markets with higher informality rates. Firm wage-setting practices may thus help explain how macroeconomic stagnation can reverse inequality reduction even when the earlier compression in worker effects persists.

This paper contributes to several strands of literature. First, we show that different forces drove Argentina’s inequality decline during the 2000s and its reversal during the

¹Our administrative records do not measure firm productivity. Complementary evidence from ENDEI links higher productivity to higher firm wage premia, but covers only manufacturing firms with at least ten employees in 2010–12. It therefore cannot establish how productivity evolved among the smallest firms.

²We measure informality in 1995–99 using household survey data and use it as a marker of weak enforcement. Firm wage effects cover registered employment only. Informality may also capture other market characteristics, and the estimates do not directly identify evasion or transitions into informal employment.

2010s: worker effects account for most of the decline, whereas firm effects more than account for the subsequent increase. Relatedly, [Alvarez et al. \(2018\)](#) show that firms contributed to the decline in inequality in Brazil during 1996–2012, and [Messina and Silva \(2019\)](#) document that two-thirds of the decline in wage variance during the 2000s occurred within skill groups across several countries in the region, suggesting a role for firm-level factors. [Haanwinckel \(2025\)](#) develops a framework linking worker composition, labor demand, and minimum wages to firm premia and sorting, and finds that demand shifts and minimum-wage increases drove Brazil’s inequality decline during 1998–2012. We build on these findings by tracking firm wage effects across both the boom of the 2000s and the stagnation of the 2010s.³ The earlier decline is consistent with evidence emphasizing educational attainment and changing returns to skills (e.g., [Lustig et al. 2016](#); [Fernández and Messina 2018](#); [Acosta et al. 2019](#)). Our findings suggest that widening differences in wage premia may help explain why inequality reduction subsequently stalled or reversed in the region.

Second, we contribute to the literature on industry wage differentials and labor market institutions by examining how they shaped firm wage inequality across contrasting macroeconomic regimes (e.g., [Krueger and Summers 1988](#); [DiNardo et al. 1996](#); [Katz and Autor 1999](#)). Following [Card et al. \(2024\)](#), we construct industry wage differentials—to our knowledge, the first application of their approach in a developing country—and show that widening gaps between sectors account for about half of the increase in firm-effect variance during the 2010s.⁴ The standard deviation of industry premia is more than twice that in the USA, while their relationship with mean worker effects is substantially weaker. These findings suggest differences in labor market structure between developing and developed economies.

We also provide new evidence on how wage floors negotiated in collective bargaining affect the dispersion of firm wage effects. Research on the “wage cushion”—the gap between actual wages and negotiated wage floors—shows how firms retain wage-setting flexibility under collective bargaining ([Cardoso and Portugal 2005](#); [Card and Cardoso 2022](#); [Bhuller et al. 2022](#)). In particular, [Card and Cardoso \(2022\)](#) show that adjustments in wage cushions partly offset increases in negotiated floors.⁵ Recent work also examines variation in firm wage effects across unions ([Derenoncourt et al. 2025b](#)) and the effects of minimum wages on firm

³[Lachowska et al. \(2023\)](#) also document cyclical variation in firm effects in Washington State, with greater dispersion during the Great Recession.

⁴In related work, [Haltiwanger et al. \(2024\)](#) show that most of the increase in USA earnings inequality between 1996 and 2018 occurred between sectors. For Italy, [Briskar et al. \(2024\)](#) also document a substantial between-industry contribution to rising earnings inequality.

⁵For Argentina, [Hermo \(2025\)](#) uses export-demand shocks to bargaining units to study how negotiated wage floors transmit shocks to firms’ wages and employment. Other papers in the Argentine context study collective bargaining coverage and wage dispersion ([Lombardo and Martinez-Correa 2019](#)), the effects of negotiated wage floors on wages, employment, and inequality ([Ramírez Leira et al. 2025](#)), and the employment effects of national minimum-wage increases ([Abbate and Jiménez 2026](#)).

premia (Bergolo et al. 2025). We complement these studies by using newly digitized data on wage floors and bargaining units’ differential exposure to a common minimum-wage increase to identify how negotiated floors compress within-industry dispersion in firm wage effects.

Finally, by studying the drivers of firm wage effects in Argentina we contribute to a growing literature quantifying firm-based inequality in developing countries within the AKM framework. Studies for, e.g., South Africa (Bassier 2023), Mexico (Pérez Pérez and Nuño-Ledesma 2024), and Uruguay (Bergolo et al. 2025) document a substantial role for firm wage premia in earnings inequality, often larger than in developed countries, a pattern that Kline (2024) signals as meriting further investigation. Our evidence identifies small firms as central to Argentina’s increase in firm-effect dispersion and links wider industry wage differentials to sectoral productivity.⁶ Furthermore, weaker institutions and regulations may allow for greater firm wage dispersion, as the increase in the variance in firm wage effects in the 2010s coincided with erosion of the real minimum wage and stagnation of negotiated floors.⁷ Similarly, local markets with higher informality rates experienced greater growth in firm-effect dispersion, consistent with theories linking informality to productivity heterogeneity (Levy 2008; Ulyssea 2018; Haanwinckel and Soares 2021).⁸

The rest of the paper is organized as follows. Section 2 describes the context and data, showing suggestive evidence that justifies our focus on firms and industries. Section 3 describes our empirical strategy. Section 4 presents our main AKM decomposition results, and Section 5 explores the drivers of changes in firm wage effects. Section 6 concludes.

2 Context and Data

This section describes the Argentine context and data used in our analysis. First, we document Argentina’s macroeconomic and institutional context in 1995 through 2019. Second, we characterize the evolution of earnings inequality in this period and discuss the role of firms in shaping these trends. Finally, we describe the data sources used in our analysis.

⁶These findings connect with a classical literature in development economics concerned with the effects of productivity differences across sectors on labor markets and growth (see, e.g., Prebisch 1950; Lewis 1954; Furtado 1963), which remains relevant today (e.g., McMillan and Rodrik 2011).

⁷While we note a weakening of the national minimum wage, several papers study the previous increase during the 2000s boom, typically finding positive effects on earnings and limited effects on employment (e.g., Maurizio and Vazquez 2016; Engbom and Moser 2022; Derenoncourt et al. 2025a).

⁸Minimum wages can also affect informal labor markets. Pérez (2020) and Derenoncourt et al. (2025a) show positive effects on informal wages in Colombia and Brazil, respectively. Machado Parente et al. (2026) show that adjustments through informality can offset formal-sector compression and raise overall inequality.

2.1 Macroeconomic and Institutional Context

Argentina experienced three distinct macroeconomic regimes during our study period, each with different implications for labor market structure and earnings inequality. The market-oriented regime of the 1990s ended in the deep 2001–02 crisis. A rapid post-crisis recovery and expansion followed, before giving way to macroeconomic stagnation in the 2010s. As shown in Figure 1, real output grew rapidly following the crisis, but fluctuated around a flat trend after 2011 while inflation accelerated. The 1990s featured trade liberalization and structural reforms that favored capital-intensive sectors. The post-crisis expansion combined a depreciated exchange rate that favored manufacturing, favorable terms of trade, and active labor market policies. Exchange-rate controls adopted in 2011 coincided with the beginning of the subsequent stagnation. Appendix Figure 1 shows that the unemployment rate fell sharply from its high levels around the crisis, while informality remained high throughout the period and also peaked around the 2001–02 crisis.

The macroeconomic regimes coincided with changes in three potential determinants of earnings inequality. First, labor productivity fluctuated and became more unequally distributed across sectors, potentially affecting wage setting through differences in sectoral productivity. Panel A of Figure 2 shows that aggregate labor productivity increased substantially during the post-crisis expansion and declined during the 2010s. Panel B shows that the employment-weighted standard deviation of log labor productivity across sectors fell during the expansion and rose steadily after 2012, from 0.588 in 2004–08 to 0.640 in 2015–19, an increase of 9.0%.⁹ Second, the skill premium varied substantially, suggesting an important role for worker characteristics, especially during the 1990s and 2000s. As shown in Panel C, a large skill premium in the 1990s declined substantially during the 2000s and then stabilized during the 2010s.¹⁰ Third, the real minimum wage increased in the 2000s, whereas collective bargaining wage floors also rose into the early 2010s, with both stagnating after 2011, as shown in Panel D.¹¹ Thus, the post-crisis expansion combined rapid growth and productivity gains with strengthening labor market institutions, whereas the 2010s combined macroeconomic stagnation with a gradual erosion of wage-setting institutions.

⁹This widening is concentrated in the lower half of the distribution: the P50–P10 gap in log productivity rose from 0.545 to 0.729, while the P90–P50 gap fell slightly from 0.838 to 0.770. [Schteingart et al. \(2022\)](#) show fluctuations of the industry wage premia in Argentina, and [Ciaschi et al. \(2021\)](#) discuss these trends in Latin America more broadly.

¹⁰The increase in the skill premium in the 1990s has been linked to the trade liberalization pursued in that decade, which favored capital-intensive sectors (e.g., [Galiani and Sanguinetti 2003](#); [Bustos 2011](#)). On the other hand, the 2000s decline is often attributed to the increase in the supply of skilled workers (e.g., [Hermo 2017](#)). See, e.g., [Manacorda et al. \(2010\)](#), [Fernández and Messina \(2018\)](#), and [Acosta et al. \(2019\)](#) for studies of the skill premium in Latin America more broadly.

¹¹The 2000s were also characterized by active policies that aimed at formalizing informal workers (e.g., [Beccaria et al. 2015](#)).

2.2 Trends in Earnings Inequality

We document trends in earnings inequality using data from social security records and the national household survey.

Earnings inequality trends. The evolution of earnings inequality in Argentina between 1995 and 2019 mirrors the macroeconomic cycle. Panel A of Figure 3 shows that the variance of log earnings rose during the 1990s, fell sharply after the 2001–02 crisis, and increased again during the 2010s stagnation. This broad pattern is common across Latin America (e.g., Gasparini et al. 2016; Messina and Silva 2019; Alvaredo et al. 2025). Appendix Figure 3 shows that these trends are similar among formal workers in the survey data and extend to informal workers as well, although with a larger magnitude.¹²

A simple decomposition suggests an important role for firms. Panel B of Figure 3 decomposes the variance of log earnings into between- and within-firm components.¹³ Approximately two-thirds of the variance of log earnings reflects differences *between* firms, and the between-firm component drives changes over time: it declined during the 2000s compression and returned to 1990s levels during the 2010s stagnation.

Role of industries. Industry differences account for nearly half of between-firm earnings variance. Panel C of Figure 3 further decomposes the between-firm component into a between-industry and a within-industry between-firm component, using 4-digit industry codes, showing that nearly half of the between-firm variance can be attributed to differences *between industries*.¹⁴ Appendix Figure 2 illustrates this divergence: by the 2010s, average monthly pay in mining and quarrying was roughly twice its 1990s level, while pay in hotels and restaurants had returned to that level after rising in the 2000s. It is also consistent with the widening dispersion of sectoral labor productivity shown in Panel B of Figure 2.

Contributions vary substantially across 4-digit industries. Appendix Figure 4 displays the eight largest and smallest 4-digit industry contributions in each period.¹⁵ Consistent with the divergence in broad-sector pay described above, oil and gas services and non-ferrous metal

¹²The survey data show stagnation rather than a clear increase in the 2010s, likely due to under-reporting in the tails of the distribution (Albina et al. 2024).

¹³To derive the decomposition, let y_{it} be log earnings of worker i in year t , and write $y_{it} = \bar{y}_{j(i),t} + (y_{it} - \bar{y}_{j(i),t})$, where $j(i)$ is i 's firm and $\bar{y}_{jt}$ is mean log earnings at j . Then, $\text{Var}(y_{it}) = \text{Var}(\bar{y}_{j(i),t}) + \text{E}[\text{Var}(y_{it} | j(i))]$.

¹⁴Following a similar procedure as in the previous footnote, we can write $\text{Var}(\bar{y}_{j(i),t}) = \text{Var}(\bar{y}_{k(j(i)),t}) + \text{E}[\text{Var}(\bar{y}_{j(i),t} | k(j(i)))]$, where $k(j)$ is j 's industry and $\bar{y}_{kt}$ is mean log earnings in industry k in period t .

¹⁵We follow Haltiwanger et al. (2024) to compute industry contributions. Write the between-industry variance as $V_t^B = \sum_k s_{kt}(\bar{y}_{kt} - \bar{y}_t)^2$, where s_{kt} is industry k 's employment share, $\bar{y}_{kt}$ is its mean log earnings, and $\bar{y}_t$ is overall mean log earnings. It follows that $\Delta V^B = \sum_k c_k$, where $c_k = s_{k1}(\bar{y}_{k1} - \bar{y}_1)^2 - s_{k0}(\bar{y}_{k0} - \bar{y}_0)^2$ is industry k 's contribution. We compute each endpoint by pooling two years and report contributions relative to the change in total earnings variance.

mining are among the leading contributors to the 2010s expansion in inequality, alongside restaurants, construction, and education. Telecommunications, banking, and pharmaceuticals made especially large contributions to the 2000s compression, with oil and gas services and non-ferrous metal mining contributing in the opposite direction. The 2000s compression was more widespread: 206 of 302 industries contributed to the decline, compared with 157 industries contributing to the increase in the 2010s.¹⁶

Role of firm size. Unlike industries, the decomposition by firm size shows that the increase in the between-firm component in the 2010s is driven by changes *within* size bins, rather than changes in the average pay conditional on firm size. Panel D of Figure 3 decomposes between-firm variance across and within four size bins: 1–5, 6–25, 26–100, and more than 100 employees. The within-bin component accounts for most of the variance and drives its evolution. The between-bin component is small and declined modestly during the 2000s. This increase could reflect employment shifting toward smaller firms with more dispersed pay. However, we observe the opposite: employment became increasingly concentrated in large firms throughout the period. Panel A of Appendix Figure 5 shows that their employment share rose from 40.2% in 1995–99 to 47.3% in 2015–19, mainly at the expense of firms with 25 or fewer employees.¹⁷

The distribution of earnings. As a result of these trends, the 2010s inequality increase was driven by a relative decline in the lower tail of the distribution. Figure 4 shows that, while lower-tail inequality declined in the 2000s, by 2011 both the P50–P5 and P50–P10 gaps had returned to their 1990s levels, reaching even higher levels in 2015–19. Upper-tail inequality, in contrast, increased in the 1990s and declined steadily during the 2000s and 2010s, although that decline slowed by the end of the period.

Women show similar trends in overall and upper-tail inequality, with a similarly large between-firm component, as shown in Appendix Figures 6 and 7. Their lower-tail inequality, however, increased throughout the period.

The evolution of earnings inequality thus closely tracks Argentina’s macroeconomic and institutional regimes, with the 2010s increase concentrated in differences between firms and the lower tail. These patterns motivate separating firm wage premia from differences in workforce composition.

¹⁶Although the contributions are broad-based, the tails are concentrated: the eight largest absolute contributions account for 43% and 41% of gross industry contributions in the two periods, respectively.

¹⁷Panel B of Appendix Figure 5 shows that these trends are similar for formal workers in the household survey data. On the other hand, Panel C shows that the distribution of informal workers across size bins is more stable over time.

2.3 Data Sources

This subsection summarizes data sources and sample restrictions used in our analysis, including data used in previous subsections. Appendix A provides details.

2.3.1 Matched employer-employee dataset

We rely on Argentina’s matched employer-employee dataset, collected by the tax authority under the *Sistema Integrado Previsional Argentino* (SIPA) for social security purposes. The data include information on all jobs held by wage-earners in the private sector, state-owned companies, and independent public institutions (such as universities and banks) for 1995–2019. We use worker and firm identifiers to track workers across firms over time.¹⁸ We observe the industry (or economic sector) and province of the firm, as well as the age and gender of workers (though not their education level). Industry codes follow a custom classification built on the ISIC revision 3.1 classification, which we use throughout the paper.

We construct a worker-firm-year panel by averaging monthly pre-tax earnings, including base salary, bonuses, severance payments, and other pay components, after deflating them to 2010 Argentine pesos.¹⁹ We exclude jobs with earnings above the 99.9th percentile within each year to limit the influence of outliers. We focus on men aged 20 to 55, a group with high and stable employment rates (Busso and Fonseca 2015), reporting results for women in the appendix. Because hours are unobserved, we limit the influence of part-time employment by excluding jobs paying less than US\$75 per month at 2010 prices. We vary this threshold in the robustness checks. We also drop jobs lasting fewer than six months, internships, and temporary-agency workers.²⁰ We also apply additional restrictions to construct the sample used for the AKM estimation.

2.3.2 Household survey data

The *Encuesta Permanente de Hogares* (or EPH for short) covers 31 urban areas and represents over 70% of Argentina’s urban population. We use it to measure informality as the share of salaried workers without regular social security contributions (International Labour

¹⁸The data do not include establishment identifiers. However, following the literature (e.g., Alvarez et al. 2018), we prefer to use firms, as wage-setting decisions are likely made at this level.

¹⁹The CPI was obtained from the National Institute of Statistics and Censuses (INDEC), except for 2007–2015 when we use alternative sources, as the official data were manipulated (see, e.g., Cavallo 2013). The alternative sources include the CPI data from several provincial statistics offices in the first years and an alternative inflation measure computed by the opposition parties in the later years of this period.

²⁰Workers in temporary employment agencies are recorded twice, appearing both as employees of the agency and as employees of the firm where they are assigned. Drenik et al. (2023) use this feature of the Argentine data to study wage-setting practices of firms hiring outsourced workers.

Organization 2014).²¹ As a complementary marker of weak enforcement of labor regulations, we also compute the share of salaried workers earning below the hourly minimum wage.²²

Earnings reported in the survey data differ from those in the administrative data for two main reasons: differences in the definition of earnings and misreporting. The EPH surveys after-tax wages while SIPA registers pre-tax wages. However, the two sources show large differences in wage levels that exceed what can be attributed to taxes and social security contributions. This is a consequence of misreporting, especially in the tails of the distribution where under-reporting is more frequent (Albina et al. 2024).

2.3.3 Minimum wage and negotiated wage floors

We compile a panel of average wage floors negotiated in sectoral collective agreements (*convenios colectivos de trabajo de actividad*) and the national minimum wage, both expressed in 2010 Argentine pesos. We obtain the national minimum wage from the Secretary of Labor, and the negotiated wage floors from a separate set of recurring Secretary of Labor resolutions that set severance-payment caps for each collective agreement.²³ Because the resolutions are not reported in a consistent manner, constructing a consistent panel of wage floors requires careful cleaning. We harmonize the names of union federations, unions, employer associations, and firms, correct data-entry typos, and reconstruct persistent bargaining units that can be followed over time even when the underlying agreement codes or signatories change.²⁴ This panel extends Hermo (2025) beyond the exporting sector and back to 1995 by digitizing wage floors from earlier resolutions.

The result is a monthly panel of average real wage floors at the level of the bargaining unit, which is illustrated alongside the real minimum wage in Panel D of Figure 2.²⁵ Additionally, we assign bargaining units to one or more 2-digit ISIC revision 3.1 industries using large language models. We manually validate the assignments and retain multiple industry codes when a unit spans industries or its coverage cannot be narrowed further.

²¹We exclude self-employment, which accounts for a quarter of the labor force according to the EPH data.

²²The Secretary of Labor publishes the monthly minimum wage, which we convert into an hourly rate by dividing it by 200 hours. Appendix A.2 describes how we compute hourly earnings in the survey.

²³The resolutions report the average wage floor within each collective agreement because it is used to compute the severance payment cap, and have been published by the Secretary of Labor since the 1990s.

²⁴Agreement codes and signatories may change over time in the raw data. Bargaining units fully renegotiate their agreements, which leads to new codes. Defining bargaining units by their signatories allows us to abstract from changes in agreement codes. However, the raw data sometimes omit signatories. We use manual review and automated matching to reconcile incomplete or inconsistent records.

²⁵An average wage floor is the mean across the wage floors of the worker categories within a bargaining unit. Because the differences across categories are typically stable, this average tracks the evolution of negotiated wages well. We replace floors that fall below the national minimum wage (MW) with the MW, as the MW is binding in those cases. This replacement has bite only for lower-wage bargaining units between 2003 and 2009, when the MW increased rapidly and the negotiated floors lagged behind.

2.3.4 Other data sources

We use two sources of productivity data. First, we obtain data on value added per worker for a sample of firms in 2010–12 from the first wave of the *Encuesta Nacional de Dinámica de Empleo e Innovación* survey (or ENDEI for short), collected by the Secretary of Labor. It covers over 3,000 manufacturing firms and can be linked to SIPA to relate productivity to firm wage premia. Second, we use data on sectoral productivity, collected and harmonized by the *Centro de Estudios para la Producción* (or CEP XXI for short), covering sectoral groups based on ISIC Rev. 3.1 during 2004–2019.

3 Methodology

The evidence in the previous section suggests that firm-specific factors play a large role in shaping earnings inequality in Argentina. To estimate the contribution of firms to inequality, we fit two-way fixed effects models to the universe of formal employment relationships in Argentina separately for five subperiods: 1995–99, 2000–04, 2005–09, 2010–14, and 2015–19. This section describes the estimation strategy and the implementation of the AKM model, the variance decomposition and bias-correction methods, and the empirical analyses used to study industry productivity, negotiated wage floors, and informality.

3.1 Firms, Workers, and the AKM Variance Decomposition

Our analysis is based on the two-way fixed effects model pioneered by [Abowd et al. \(1999\)](#):

$$y_{it} = \alpha_i + \psi_{j(i,t)} + \Upsilon_t + \varepsilon_{it}, \quad (1)$$

where y_{it} is log earnings and $j(i, t)$ identifies worker i 's employer in year t . Worker effects α_i capture attributes rewarded equally across employers within each subperiod, while firm effects ψ_j capture wage premia common to a firm's workers. Year effects Υ_t absorb common earnings fluctuations, and ε_{it} is the residual. We include no additional worker covariates.

Firm wage effects are identified from workers moving across employers. We assume exogenous mobility: the residual is mean independent of a worker's entire employer history, conditional on worker and year effects ([Card et al. 2013](#); [Kline 2024](#)):

$$E[\varepsilon_{it} \mid \alpha_i, \{j(i, s)\}_{s \in T}, \Upsilon_t] = 0 \text{ for all } t \in T,$$

where T is the set of years in each estimation sample. Under this assumption, expected earnings changes for movers identify differences in firm premia after accounting for year

effects. The assumption permits sorting on worker and firm effects but rules out mobility systematically related to residual earnings. We present diagnostic evidence on the plausibility of this assumption in the next section.

Variance decomposition. Equation (1) allows us to decompose the variance of log earnings into the following components:

$$\begin{aligned} \text{Var}(y_{it}) = & \underbrace{\text{Var}(\alpha_i)}_{\text{Worker}} + \underbrace{\text{Var}(\psi_{j(i,t)})}_{\text{Firm}} + \underbrace{2 * \text{Cov}(\alpha_i, \psi_{j(i,t)})}_{\text{Sorting}} \\ & + \underbrace{\text{Var}(\Upsilon_t) + 2 * \text{Cov}(\alpha_i, \Upsilon_t) + 2 * \text{Cov}(\psi_{j(i,t)}, \Upsilon_t)}_{\text{Covariates}} + \underbrace{\text{Var}(\varepsilon_{it})}_{\text{Residual}}. \end{aligned} \quad (2)$$

Plug-in estimates of these variance components can exhibit substantial limited mobility bias (see, e.g., [Andrews et al. 2008](#); [Bonhomme et al. 2023](#)). Our baseline uses the heteroskedasticity-robust correction of [Kline et al. \(2020\)](#), or KSS for short. We also report uncorrected estimates and estimates using the homoskedastic correction of [Andrews et al. \(2008\)](#).

Connectedness and implementation. Firm wage effects can only be identified for a set of firms that are connected by worker mobility. Our baseline analysis uses the largest set that remains connected after removing any single worker-firm match, as required by the KSS correction. We also report plug-in estimates for the largest connected set. For granular analyses where bias correction is infeasible, we use plug-in estimates and discuss the implications.²⁶ We explicitly note when each approach is used in the analysis. Throughout this paper, we use the open-source `pytwoway` Python package to implement the estimation and bias correction methods ([Bonhomme et al. 2023](#)).

Analysis sample for the AKM estimation. We apply additional sample restrictions to the data. Following common practice in the AKM literature, we keep the highest-paying job per worker-year when workers have multiple jobs in the same year. We drop single stayers before constructing the two connected sets. Table 1 compares the full sample with the two connected sets. The leave-one-out set retains 73.7% of worker-years in 1995–99 and 78.8% in 2015–19. The leave-one-out connected set drops proportionally more firms than workers, suggesting that the excluded firms are smaller on average. Despite this selection toward larger firms, earnings and age distributions remain similar to those in the full sample.

²⁶The KSS bias correction method requires estimation of separate models for each group, which is computationally intensive and may not be feasible for very granular groups with low mobility. See [Azkarate-Askasua and Zerecero \(2024\)](#) for a related discussion.

Variance decomposition by region and size. For each region and size group, we construct the connected sets and estimate a separate AKM model with the KSS correction. The groups are defined as follows. First, we group provinces into six regions based on geographical proximity: “Buenos Aires,” “Center,” “West,” “Northwest,” “Northeast,” and “Patagonia”.²⁷ Second, we define three size bins based on the average number of employees within each sub-period: small (25 or fewer employees), medium (more than 25 and at most 100 employees), and large (more than 100 employees).²⁸

Validation of estimated firm wage effects. To validate our estimates, we relate 2010–14 firm effects to log value added per worker in the 2010–12 ENDEI survey. Regressions control for log firm size, its interaction with log productivity, and province and industry fixed effects, weighting firms by worker-years.

3.2 The Drivers of Firm Wage Effects

We explore three potential drivers of firm wage effects. First, motivated by the growing dispersion in productivity between industries, we study the role of industry differences in accounting for the evolution of the firm component. Second, as the strength of labor market institutions fluctuated significantly during our study period, we study the effect of changes in wage floors on the distribution of firm wage effects. Finally, given that the 2010s inequality increase seems to be concentrated among smaller firms and in the lower tail of the earnings distribution, we study the association between the informality rate, as an indicator of weak enforcement of labor regulations, and the evolution of firm wage effects.

3.2.1 Industry wage differentials

To assess whether between-industry wage differences reflect firm wage premia rather than workforce composition, we construct industry wage differentials following [Card et al. \(2024\)](#). These are defined as averages of firm effects within 4-digit industries, weighted by worker-

²⁷The regions are defined as follows. Buenos Aires: City of Buenos Aires and Buenos Aires Province; Center: La Pampa, Córdoba, Santa Fe, and Entre Ríos; West: Mendoza, San Juan, La Rioja, and San Luis; Northwest: Jujuy, Salta, Tucumán, Catamarca, and Santiago del Estero; Northeast: Formosa, Chaco, Misiones, Corrientes; Patagonia: Neuquén, Río Negro, Santa Cruz, Chubut, Tierra del Fuego.

²⁸For evidence on the employer-size wage premium and its evolution see, e.g., [Brown and Medoff \(1989\)](#) and [Bloom et al. \(2018\)](#).

years.²⁹ Formally,

$$\psi_{kp} = \frac{\sum_{k(j)=k} N_{jp} \psi_{jp}}{\sum_{k(j)=k} N_{jp}} \quad (3)$$

where ψ_{jp} is the (estimated) firm fixed effect for firm j in period p , $k(j)$ is the industry of firm j , and N_{jp} is the number of worker-year observations for firm j in period p . As in [Card et al. \(2024\)](#), we normalize mean firm effects in restaurants (ISIC 5521) to zero within each period. The weighted variance of industry differentials measures the between-industry component of firm-effect variance and does not depend on this normalization.

Sectoral productivity. We expect the industry wage differentials to reflect differences in mean labor productivity across sectors. To test this relationship, we rely on data on average labor productivity from national accounts at a more aggregate level than our industry classification.³⁰ We construct industry wage differentials at that level and correlate them with the average labor productivity within each period.

We also examine whether less productive industries exhibit greater dispersion in firm wage effects. Distortions that weaken the allocation of resources toward more productive firms can both lower average productivity and sustain productivity dispersion within an industry ([Hsieh and Klenow 2009](#)). Differences in regulatory compliance, including unregistered employment and unreported wage payments, may provide another source of heterogeneity across firms ([Ulyssea 2018](#); [Feinmann et al. 2022](#)). We also compute the within-sector variance of firm effects at the same aggregation level and relate it to sectoral productivity.

We emphasize that, as the variation in productivity across industries is not exogenous to changes in the mean and variance of firm wage effects, these analyses do not establish causality. We interpret the results as offering suggestive evidence on the role of labor productivity in accounting for firm-based inequality.

3.2.2 Labor market institutions

The national minimum wage and negotiated wage floors more than doubled between the late 1990s and late 2000s. We examine whether industries with larger increases in negotiated wage floors experienced greater compression in firm wage effects.

²⁹A different approach to study wage differences across industries is to estimate industry fixed effects directly (e.g., [Krueger and Summers 1988](#)). [Card et al. \(2024\)](#) show that such an approach suffers from “hierarchy bias,” as movers from low-wage to high-wage industries tend to move from relatively good firms in the origin industry to relatively bad firms in the destination industry, thereby biasing estimates of industry fixed effects. The “bottom-up” approach of [Card et al. \(2024\)](#) bypasses this problem by aggregating the unbiased estimates of firm wage effects.

³⁰The labor productivity data are available at an aggregate sectoral definition that generally corresponds to ISIC sections and is more disaggregated within some sections, such as manufacturing.

We use the newly digitized collective bargaining data to examine the relationship between changes in negotiated wage floors and the distribution of firm wage effects within ISIC subsections. We measure wage-floor growth as the change in the mean log wage floor between 2001–02 and 2008–09, the period of the largest increase in wage floors.³¹ We estimate its effect on changes in the dispersion and percentiles of log earnings, firm effects, and worker effects from 1995–99 to 2005–09 and from 1995–99 to 2015–19.

Variation in negotiated wage floors may be endogenous to changes in these outcomes. To estimate their causal effect, we instrument the change in the mean log wage floor with the negative gap between the industry’s mean log wage floor in 2001–02 and the average log national minimum wage in 2005–07. The exclusion restriction requires that outcomes in the initially lower wage-floor industries would have evolved similarly to those in initially higher wage-floor industries in the absence of the minimum-wage increase. Limited adjustment of negotiated wage floors during the late 1990s and early 2000s, before the sharp minimum-wage increase, motivates this design. We further assess the validity of the instrument using pre-trend evidence.

3.2.3 Informality and firm wage inequality

Under weak enforcement, firms may respond to adverse conditions by hiring informally or paying below the minimum wage (Ponczek and Ulyssea 2021). Motivated by the concentration of rising inequality among small firms and at the bottom of the distribution, we examine whether markets with higher initial informality rates experienced larger increases in firm-effect dispersion. We use the 1995–99 informality rate as a marker of weak enforcement, while recognizing that it may capture other differences across local labor markets.

We aggregate our estimated firm and worker effects to local labor markets, defined as the intersection of the seven regions and the sections of ISIC revision 3.1, weighting by the number of worker-years in each firm.³² We measure informality and compliance with the minimum wage in each market using the survey data, as described in Section 2.3.2. As the KSS bias correction is not feasible at this level of granularity, we rely on the plug-in estimator of the variance of firm and worker effects.³³

³¹The wage-floor measures use bargaining units that can be followed consistently from the late 1990s through 2008, and we restrict the sample to industries with at least two wage-floor observations in 2001–02. The average number of wage floors per industry is 11.2, with a minimum of 2 and a maximum of 48.

³²Some ISIC sections are combined to match the aggregation level of the EPH classification, resulting in 11 economic sectors and 67 local labor markets, after dropping market-year cells with fewer than 30 observations in the EPH. To match the EPH geography, this analysis treats the City of Buenos Aires and Buenos Aires Province as separate regions, whereas the six-region classification used in the baseline variance decomposition combines them.

³³Local labor market fixed effects absorb permanent differences in bias across markets. When presenting these results, we discuss concerns about time-varying bias and provide supporting evidence from outcomes

We fit a two-way fixed effects model with local labor market and period fixed effects, interacting the informality rate in 1995–99 with period indicators.³⁴ The interaction for 1995–99 is omitted, so our parameters of interest measure the differential change in the association between the informality rate and the outcome relative to the 1995–99 baseline. We weight by the number of worker-years and cluster standard errors at the local labor market level. While this analysis does not establish causality, it characterizes the labor market conditions associated with changes in firm wage inequality.

4 Firms and the Evolution of Earnings Inequality

This section presents the AKM variance decomposition and examines heterogeneity by firm size, pay rank, and region. We then assess the identifying assumptions and robustness of the estimates and compare the results with the literature.

4.1 The AKM Variance Decomposition

The firm component accounts for a significant share of earnings inequality, particularly in the 2010s. Table 2 shows estimates of the KSS-corrected decomposition for our five subperiods. Consider the first period, 1995–99, displayed in column (1). Out of the total variance of 0.554, the worker component accounts for 0.234 (42.2%), the firm component for 0.183 (33.0%), and the sorting component for 0.096 (17.3%). The variance decomposition for 2015–19, displayed in column (5), shows that the firm component accounts for 0.181 of a total variance of 0.436 (41.5%). This increase of 8.4 percentage points is fully accounted for by a decline of 9.1 percentage points in the importance of the worker component.

Earnings inequality declined by 24.5% between 1995–99 and 2005–09, driven by both the worker and firm components. Worker effects account for 60.3% of this decline and firm effects for 30.9%. The larger worker contribution is consistent with evidence emphasizing educational upgrading and declining returns to skills.

In contrast, firm effects more than account for the subsequent increase in earnings inequality. Between 2005–09 and 2015–19, firm-effect variance rose by 0.040, more than twice the 0.018 increase in earnings variance, offsetting declines in other components. Figure 5 shows that firm-effect variance returned close to its 1990s level, while worker-effect variance remained substantially lower. By the end of our period, the firm component accounts for a larger share of earnings inequality than the worker component.

that do not rely on estimated AKM effects.

³⁴More precisely, we estimate $y_{ct} = \gamma_c + \delta_t + \sum_{s \neq 1} \beta_s I_{c1} \mathbf{1}\{t = s\} + e_{ct}$, where y_{ct} is an outcome in market c and period t , I_{c1} is the informality rate in 1995–99, and γ_c and δ_t are market and period fixed effects.

Sorting. Changes in sorting played a smaller role in the evolution of earnings inequality. Sorting accounts for an average of 17% of earnings variance, but contributes less to its changes than worker or firm effects. The slight decline in the sorting component contrasts with evidence from advanced economies, where increasing positive assortative matching has been an important driver of *rising* inequality (e.g., Card et al. 2013; Song et al. 2019).

Women. Estimates for women show similar trends to those for men. Appendix Figure 8 shows a decline in worker and firm effect variances from 1995–99 to 2005–09 and an increase in the firm effect variance between 2005–09 and 2015–19, similar to the patterns for men. However, we find a negative sorting component in all periods, which is mitigated by the variance corrections but remains slightly negative.³⁵

4.2 Heterogeneity by Firm Size, Pay Rank, and Region

We now examine how firm-effect dispersion evolved across firm sizes, pay ranks, and regions.

4.2.1 Firm size and pay rank

Firm size. The 2010s show a size gradient in the evolution of the firm component. Column (7) of Table 3 shows that between 2005–09 and 2015–19 the variance of firm effects rose by 0.085 among small firms, 0.043 among medium firms, and 0.020 among large firms. The increase among the smallest firms was more than *four times* as large as among the largest. The 2000s show no comparable gradient: the variance changed by -0.036 among small firms, -0.052 among medium firms, and -0.016 among large firms.

The 2010s increase in firm-effect dispersion was concentrated among small firms, whose declining employment share partly offset the rise in aggregate dispersion. To distinguish these changes from shifts in average premia across size groups, we decompose firm-effect variance into between- and within-size components.³⁶ Panel B of Table 3 shows that within-size dispersion accounts for 80.5% of the increase between 2005–09 and 2015–19. Panel C shows that the share of worker-years in small firms declined by 4.6 percentage points over this period, while that in large firms increased by 5.6 points. Employment thus shifted away

³⁵Appendix Table 1 replicates the sample statistics in Table 1 for women. On average, the number of worker-years in the connected set is 62.8% smaller than for men, and the share with more than one firm is 17.2% smaller, raising stronger concerns about limited mobility bias.

³⁶More precisely, for each period we estimate the variance-corrected $\text{Var}(\psi_j \mid j \in b)$ separately for each bin b . Then, using the full male leave-out-match sample for that period, we compute the worker-year weights w_b and the (weighted) mean firm effects ψ_b for each bin. The within-bin component is $\sum_b w_b \text{Var}(\psi_j \mid j \in b)$ and the between-bin component is $\sum_b w_b (\psi_b - \psi)^2$, where $\psi = \sum_b w_b \psi_b$ is the overall (weighted) mean firm effect. The sum of the within- and between-bin components equals the total variance reported in the table.

from the size groups with the largest increases in dispersion. Between-size differences played a larger role in the earlier decline than in the subsequent increase (Panel B), consistent with compression of the wage-size gradient during the 2000s.

Percentiles of firm wage effects. Appendix Figure 9 shows the evolution of the lower- and upper-tail gaps of the firm effect distribution.³⁷ Lower-tail gaps—P50–P5, P50–P10, and P50–P25—compressed substantially during the 2000s. This compression partially reversed in the 2010s. By 2015–19, the P50–P5 gap exceeded its 1995–99 level, while the P50–P25 gap did not increase during the 2010s, showing that the 2010s increase in the dispersion of firm effects was concentrated among the lowest-paying firms. Upper-tail gaps remained relatively stable, with a slight increase in 2000–04 and decline in 2010–14, relative to 1995–99. These findings parallel the lower-tail increase in earnings inequality documented in Section 2.2.

4.2.2 Geographic variation.

The increase in firm-effect dispersion was broad-based across regions. Differences in average firm effects between regions account for only 6.6% of the variance on average, implying that most of the dispersion reflects differences within regions. Furthermore, within-region dispersion accounts for 92.4% of the increase between 2005–09 and 2015–19.

Within this broad-based increase, a regional gradient emerges. Appendix Figure 10 shows that less developed regions have substantially higher levels of firm-effect dispersion throughout the sample, particularly the Northwest, the Northeast, and the West, whereas Buenos Aires and Patagonia display lower levels. The increase in the variance of firm effects in the 2010s is somewhat larger in lower-income regions, ranging from 0.033 to 0.092 across regions.

The patterns discussed in this section motivate our examination of productivity, wage-setting institutions, and informality as potential drivers of the reversal.

4.3 Robustness and Validation

We assess the AKM identifying assumptions, examine robustness to alternative estimators and sample definitions, and relate the estimated firm effects to firm productivity.

4.3.1 Testing AKM assumptions

We present three diagnostic exercises supporting the AKM specification. First, exogenous mobility implies parallel wage trends before moves across different origin and destination

³⁷This figure relies on the largest connected set of firms and the plug-in variance estimator.

firms. Following Card et al. (2013), we classify the mover’s origin and destination firms into quartiles based on their co-workers’ wages and study their wage evolution around the time of the move.³⁸ Appendix Figure 11 shows the evolution of wages for workers moving across firms to and from the first and fourth quartiles only, for the intervals 1997–01, 2006–10, and 2013–17. The evolution of wages before the move is consistent with exogenous mobility: wages evolve in parallel before the move regardless of the origin and destination firms.

Second, under exogenous mobility, the model predicts equal and opposite wage changes for moves between the same pair of firms.³⁹ Appendix Figure 11 shows approximately symmetric wage gains and losses for moves between low- and high-wage firms. Appendix Figure 12 extends this comparison to all quartile pairs. A slight deviation occurs between the first and fourth quartiles, where upward movers experience slightly larger gains than the losses of downward movers. Overall, we consider the deviations from symmetry to be small.

Finally, we assess the additive specification by computing mean residuals within a 10-by-10 grid of worker- and firm-effect deciles (Card et al. 2013). Appendix Figure 13 shows that, in all cases, the mean residuals are within approximately 1.1% of zero (and are usually much smaller), suggesting that the no-interaction assumption is a good approximation.⁴⁰

4.3.2 Robustness checks

Alternative estimators for the variance decomposition. The KSS correction requires a more restrictive connected set that typically excludes smaller firms with less mobile workers. We therefore report the plug-in decomposition for the largest connected set in Appendix Table 2 and compare estimators holding the leave-one-out sample fixed in Appendix Figure 14. The plug-in estimator yields a larger firm component and a smaller sorting component, but the evolution of firm-effect dispersion is similar across samples and estimators.

Varying minimum earnings cutoff. Our baseline imposes a minimum earnings cutoff of USD 75 in 2010. Appendix Figure 15 reports plug-in estimates for cutoffs of USD 100, 125, 150, and 200. Higher cutoffs reduce the level and relative importance of the firm component, but not its evolution: the worker component explains most of the decline from 1995–99 to

³⁸To prevent compositional changes over time we focus on movers that are observed during four consecutive years around the move. We residualize wages by year fixed effects to adjust for the macroeconomic cycle.

³⁹To see this, consider workers moving from firm j to firm k . Under exogenous mobility their expected wage change should be $\psi_k - \psi_j$ (up to the year fixed effects). Similarly, workers moving from k to j should experience the reverse wage change, $\psi_j - \psi_k$.

⁴⁰Similar to other papers in this literature (e.g., Alvarez et al. 2018), we find the largest deviations for the low-wage workers. The mean residuals are positive for workers in the lowest deciles of worker and firm fixed effects in 1995–99 and 2000–04, but turn negative in the subsequent periods, which may be related to the evolution of the minimum wage.

2005–09, while the firm component accounts for the increase from 2005–09 to 2015–19.

Time-varying five-year window. Appendix Figure 16 shows that rolling five-year estimates confirm the timing of the reversal. The firm component falls sharply between the 2001–05 and 2004–08 windows, and increases steadily thereafter, returning to its 1990s level. The compression coincides with the sharp minimum-wage increases of the 2000s. The worker component declines more gradually, between the 1999–03 and 2007–11 windows, and subsequently stabilizes, paralleling the evolution of returns to education.

4.3.3 Estimated firm wage effects

We validate our estimated firm wage effects against firm productivity. We match data on value added per worker for a sample of over 3,000 manufacturing firms in 2010–12 to the estimated firm effects from the 2010–14 period. Appendix Table 3 shows results of regressions of firm effects on log value added per worker, log firm size, and their interaction, with and without controls for industry and province.

Productivity is positively associated with firm wage premia across specifications. Evaluated at the mean log firm size across sampled firms (corresponding to approximately 55 employees), the specifications with size interactions imply that a 10% increase in value added per worker is associated with 0.8–1.0% higher wages holding worker characteristics fixed. The association weakens with firm size, particularly when controlling for both industry and province.⁴¹ In sum, more productive firms pay higher wages, holding worker characteristics fixed, consistent with rent-sharing between firms and workers (Card et al. 2018).

4.4 Comparison with the Literature

We compare the contribution of firm wage effects in Argentina with estimates from developing and advanced economies. We rely on estimates using the plug-in variance estimator, as most previous studies do. All specifications control for year fixed effects, some also control for either age or education, and all but one focus on a sample of males. Time windows overlap with ours but are generally shorter.

The share of variance explained by firms in Argentina is substantial in the international context. In 2015–19, this share exceeds those reported for the other countries shown in the left panel of Appendix Figure 17, including developing economies like Brazil (Alvarez

⁴¹Alvarez et al. (2018, Table 6) run the analogous regression for the universe of Brazilian manufacturing firms. In 2008–12, the period closest to ours, they report coefficients on log value added per worker of 0.140 without controls and 0.120–0.128 with subsector or state controls, close to our estimates from a much smaller survey-based sample. Their interaction with firm size is positive.

et al. 2018), Mexico (Pérez Pérez and Nuño-Ledesma 2024), South Africa (Bassier 2023), and Uruguay (Bergolo et al. 2025), as well as advanced economies such as West Germany (Card et al. 2013) and the USA (Song et al. 2019).

However, as argued by Kline (2024, p. 6), comparing variance levels may be more appropriate than shares due to potential differences in “intrinsic noise levels” across datasets. When comparing levels, Argentina’s firm-effect variance remains larger than in advanced economies but is more comparable to other developing countries. In particular, the variance of the firm wage effects is similar to that of Mexico (Pérez Pérez and Nuño-Ledesma 2024) and smaller than that of South Africa (Bassier 2023). The key difference between Argentina and other developing countries seems to be in the smaller variance of worker effects.

5 Potential Drivers of Firm Wage Inequality

The preceding analysis shows that firm wage premia compressed during the 2000s expansion and became more dispersed during the stagnation of the 2010s. Motivated by the descriptive evidence in Section 2, we explore three potential mechanisms that may have contributed to these changes. Productivity dispersion across sectors increased in the 2010s, wage-setting institutions strengthened in the 2000s and weakened in the 2010s, and the 2010s inequality increase was concentrated at the bottom of the wage distribution.

We first quantify the contribution of industry factors and relate firm wage inequality to sectoral productivity. We then examine whether rising negotiated wage floors contributed to within-industry compression during the 2000s, and whether this effect persisted into the 2010s. Finally, we combine administrative records with household survey data to examine whether markets with higher initial informality experienced greater increases in firm wage inequality and shifts toward smaller firms, particularly during the 2010s stagnation.

5.1 Industries, Productivity, and Firm Wage Inequality

We construct industry wage differentials and quantify their contribution to changes in the variance of firm effects, relate these differentials to sectoral labor productivity, and compare the distribution of industry premia in Argentina and the USA.

5.1.1 Industry wage differentials and their contribution

Following Card et al. (2024), we measure industry wage differentials by averaging the estimated firm wage effects within 4-digit industries, weighting firms by worker-years. Panel B of Table 4 shows that the between-industry component accounts for 38.4% of the firm-effect

variance in 1995–99 (column 1), and increases to 41.1% in 2015–19 (column 3). While these results are based on the plug-in variance estimator, the table shows that they capture the same evolution of the variance of firm effects as the KSS estimator used in Section 4.1.

Contribution to the evolution of firm wage inequality. Industries account for around half of the increase in the variance of firm wage effects during the 2010s. Panel B of Table 4 shows that the variance of firm effects rose from 0.150 in 2005–09 to 0.191 in 2015–19. The between-industry component increased from 0.060 to 0.078, accounting for 45.2% of the rise. By comparison, between-industry differences account for about a third of the earlier decline (column 4), pointing to a larger role for within-industry compression during the 2000s.

The industry rankings broadly align with the raw-earnings patterns in Section 2.2. In 2015–19, industries with the highest premia include non-ferrous metal mining, motor vehicle manufacturing, services incidental to oil and gas extraction, and banking, and those with the lowest premia include education, construction-related activities, used-book retail, and fishing services. Several industries at both ends of this ranking also contributed substantially to the 2010s increase in between-industry earnings inequality in the raw data: oil and gas services and non-ferrous metal mining among high-premium industries, and construction and education among low-premium industries (Appendix Figure 4). Appendix Table 4 reports the ten highest and lowest wage differentials at the 4-digit level in 2015–19, along with the mean worker effect and employment share in each industry. Appendix Table 5 reports differentials at the more aggregated ISIC section level, for 1995–99, 2005–09, and 2015–19.

Comparison with industry fixed effects. Hierarchy bias predicts attenuation of industry fixed effects estimated from industry movers, as discussed in Section 3.2.1. Consistent with this prediction, Appendix Figure 18 shows smaller industry gaps from those estimates than from averages of firm effects.

5.1.2 Industries, firm wage effects, and labor productivity

We relate industry wage differentials to labor productivity, measured as value added per worker. We construct industry wage differentials at the same level of aggregation as the productivity data. Panels A and B of Figure 6 show that more productive industries pay higher firm premia in both 2005–09 and 2015–19, consistent with industry wage differentials partly reflecting productivity differences.

Notably, less productive industries not only pay lower firm premia on average but also exhibit greater dispersion in the premia paid by their firms. Panels C and D of Figure 6 show that this negative association with within-industry dispersion strengthens over time.

A 10% increase in labor productivity is associated with a reduction in the within-industry variance of firm effects of about 0.0022 in 2005–09 and 0.0045 in 2015–19. We reject equality of the two slopes at the 5% level (p -value = 0.043).

These cross-industry patterns parallel the evolution across macroeconomic regimes. During the 2000s expansion, aggregate productivity rose and the dispersion of firm wage effects declined. During the 2010s stagnation, productivity declined and both productivity gaps and the dispersion of firm wage effects increased.⁴² While the evidence does not establish causality, the similarity between the cross-sectional and aggregate patterns is consistent with productivity playing a role in firm wage inequality.

Labor productivity is associated with the dispersion of firm effects but not of worker effects, reinforcing the interpretation that productivity may affect inequality through the firm wage-setting process. Panels A and B of Appendix Figure 19 show that more productive industries employ workers with higher mean worker effects, although this gradient is weaker than for firm effects. Panels C and D show no association with the within-industry dispersion of worker effects, with coefficients close to zero and statistically insignificant.

5.1.3 International benchmark and interpretation

Industry wage differentials are substantially more dispersed in Argentina than in the USA. Figure 7 compares the distribution of industry premia for Argentina (2015–19) with the USA estimates of Card et al. (2024) for 2010–18, both computed relative to restaurants.⁴³ The standard deviation in Argentina (0.281) is more than twice that in the USA (0.122). Most USA industries pay modest positive premia relative to restaurants, whereas Argentina’s distribution is much wider.

The industry wage differentials are correlated with mean worker effects, reflecting the sorting of workers across industries. Appendix Figure 20 shows this relationship for 1995–99, 2005–09, and 2015–19. The estimated slope ranges from 0.34 to 0.42 across periods, compared with 0.90 in the USA (Card et al. 2024, Figure 3). Thus, Argentina combines greater dispersion in industry wage premia with a substantially weaker relationship between industry premia and worker composition.

These patterns connect to development theories that emphasize productivity heterogeneity across sectors. The dual-economy tradition focuses on the reallocation of labor toward high-productivity activities (Lewis 1954), a process that has contributed relatively little, and at times negatively, to growth in Latin America (McMillan and Rodrik 2011). The Latin American structuralist tradition instead emphasizes “structural heterogeneity”: the coexis-

⁴²See Section 2.1 for productivity trends and Section 4.1 for firm-effect dispersion.

⁴³Both classifications use 4-digit industries: ISIC Revision 3.1 for Argentina and NAICS for the USA.

tence of modern, high-productivity activities with large low-productivity segments (Prebisch 1950; Furtado 1963; Pinto 1970). Although our estimates measure wage premia rather than productivity, their substantial dispersion and association with sectoral productivity provide a labor-market counterpart to this heterogeneity.

Industry differences account for nearly half of the increase in firm-effect variance during the 2010s, compared with about a third of the decline during the 2000s. These findings point to a larger role for within-industry compression in the 2000s boom. We next examine whether labor market institutions contributed to this compression.

5.2 Labor Market Institutions

The real national minimum wage and negotiated wage floors rose rapidly during the 2000s, when firm wage effects compressed, and weakened in the 2010s, as the variance of firm wage effects increased. We examine whether wage-floor growth compressed firm effects within 2-digit industries, using the sharp minimum-wage increase as a source of plausibly exogenous variation. We estimate the effects of this growth on changes from 1995–99 to 2005–09 and to 2015–19, assessing whether the initial compression persisted.

5.2.1 OLS estimates

We find a significant association between increases in the average wage floor and declines in both earnings inequality and the dispersion of firm wage effects between 1995–99 and 2005–09. Panel A of Table 5 shows that a 10 log point larger increase in the mean wage floor is associated with a decline of 0.0075 in the standard deviation (SD) of log earnings, of 0.0080 in the SD of firm effects, and of 0.0041 in the SD of worker effects (columns 1, 3, and 5, respectively), all significant at the 10% level. These correspond to 1.2%, 2.1%, and 0.8% of the respective 1995–99 levels. Notably, these associations weaken over the longer horizon. Columns (1), (3), and (5) of Panel A of Appendix Table 6 show smaller estimates that are statistically indistinguishable from zero for changes through 2015–19.

5.2.2 IV estimates

A concern with these results is that wage floors are determined by other factors that also affect earnings dispersion. For example, productivity gains concentrated among smaller firms could both narrow firm wage gaps and encourage higher negotiated floors. To address this concern, we instrument the change in the mean log wage floor with the negative gap between the industry’s mean log wage floor in 2001–02 and the average log minimum wage in 2005–07. The minimum-wage increase placed more pressure on industries whose initial floors

were relatively low. A causal interpretation requires that the initial gap affected changes in firm wage effects only through its effect on wage floors. The institutional history supports this view: wage floors changed little in the decade preceding the 2004–07 minimum-wage increase, which prompted essentially all industries, regardless of industry characteristics, to renegotiate and update their wage floors.⁴⁴

The gap instrument is strongly predictive of the growth in the negotiated wage floors. Panel A of Appendix Table 7 reports that the coefficient on the gap in the regression for the change in the mean log wage floor is 0.550 (SE = 0.077) with an F statistic of 51. Consistent with the institutional compression mechanism, the instrument predicts a larger increase in the lower percentiles of the wage floor distribution, as shown in columns (2) through (6), and a strong decline in the dispersion of wage floors, as shown in column (7).

The reduced-form relationship follows the timing of the minimum-wage increase. Panel B of Appendix Table 7 relates the initial wage-floor gap to changes in the variance of firm effects relative to 1995–99. The coefficient is small and insignificant in 2000–04, largely before the minimum-wage increase (p -value = 0.144), becomes more negative and statistically significant only in 2005–09, and subsequently diminishes and loses significance. This timing supports the interpretation that the minimum-wage increase drove the compression.

The IV estimates imply greater compression of firm effects than OLS, with gains concentrated in the lower tail. Column (4) of Table 5 shows that a 10 log point larger increase in the mean wage floor leads to a decline of 0.0127 in the SD of firm effects ($t = 1.91$), or 3.3% of its 0.382 level in 1995–99. The OLS estimate in column (3) is smaller, at 0.0080 (2.1% of the 1995–99 level), although the difference is not statistically significant. For the SD of worker effects, the IV and OLS point estimates are more similar. Panel B of Table 5 shows that a 10 log point larger increase in the mean wage floor implies approximately 4.1% higher wages at the 10th percentile of the firm-effect distribution, holding worker characteristics fixed, compared with an OLS estimate of 2.5%. The mean wage floor also increases the 25th percentile, whereas the estimate is smaller and insignificant at the 90th percentile. The estimated effects are consistent with institutional compression operating from below.⁴⁵

Similarly to OLS, the IV estimates are generally smaller for the longer-run changes between 1995–99 and 2015–19, as shown in Appendix Table 6. Panel A shows statistically

⁴⁴The lack of negotiations during the 1990s follows from the economic and political conditions of the time. Price stability, including mild deflation, and a political environment unfavorable to labor unions, together with the ultra-activity clause (which meant that expired agreements remained in force), discouraged negotiations (Palomino and Trajtemberg 2006). Collective bargaining revived in the second half of the 2000s.

⁴⁵With only 43 industries, we verify that these estimates are robust to small-sample inference. Because the model is just identified, the wild bootstrap p -value of the reduced form, computed over 9,999 replications, is the Anderson–Rubin p -value for the null of no effect (e.g., Cameron et al. 2008). The bootstrap p -values are 0.084 for the SD of firm effects, 0.039 and 0.049 for the 10th and 25th percentiles, and 0.893 for the 90th.

insignificant estimates for the SD of log earnings, firm effects, and worker effects. Panel B shows somewhat larger estimates for the 10th and 25th percentiles of firm effects than OLS, and a statistically insignificant estimate for the 90th percentile.

Economic magnitude. Our design identifies how firm-effect dispersion responded to relative differences across industries in wage-floor growth, and those differences are large: wage floors rose by 50.3 log points on average, with a cross-industry SD of 23.6 log points. The IV estimates imply that an industry whose wage floors grew one SD faster saw the SD of firm effects fall by an additional 0.030, or 7.8% of the average within-industry dispersion of 0.382 in 1995–99. This magnitude is large relative to the compression observed over the same period, when average within-industry dispersion fell from 0.382 to 0.325.

Negotiated wage floors compressed firm wage effects within industries during the 2000s macroeconomic boom. The subsequent stagnation of negotiated floors and the deteriorating macroeconomic conditions may have weakened this compressing force.

5.3 Informality and Firm-Based Inequality

Weak enforcement of labor regulations may have allowed low-paying firms to fall further behind as the macroeconomy deteriorated. Because informality directly captures noncompliance with social security registration, we use it as a marker of weak enforcement, although we acknowledge it may also reflect other differences across local labor markets. We define local markets by the intersection of regions and sectors, measure their informality rates using the EPH, and aggregate our firm effects at the same level. To abstract from changes in informality levels, we use the initial informality rate in 1995–99 throughout the analysis.

5.3.1 Informality and firm wage inequality

Firm wage inequality grew more in markets with higher initial informality, particularly in the 2010s. Column (2) of Table 6 shows that a 10 percentage point higher initial informality rate is associated with an additional 0.0121 increase in the SD of firm effects between 1995–99 and 2015–19 ($t = 2.09$). This corresponds to 3.0% of its 0.406 baseline level. The differential change is indistinguishable from zero in 2000–04, becomes positive in 2005–09, and is larger in both 2010s periods. Column (1) shows a similar pattern for earnings dispersion.

This divergence is driven by the lower tail of the firm-effect distribution. Columns (3) and (4) of Table 6 show that, between 1995–99 and 2015–19, a 10 percentage point higher initial informality rate is associated with an additional wage decline of approximately 3.0% at the 10th percentile of firm effects, holding worker characteristics fixed ($t = 3.09$). The

corresponding change at the 90th percentile is small and statistically indistinguishable from zero ($t = 0.51$). The widening of firm wage inequality in initially high-informality markets thus reflects the lowest-paying firms falling behind.

5.3.2 Informality, firm size, and minimum wage compliance

Initially high-informality markets also shifted toward smaller formal firms. Column (5) of Table 6 shows that a 10 percentage point higher informality rate is associated with a 4.3% larger decline in mean firm size between 1995–99 and 2015–19 ($t = 2.33$). Columns (1) and (2) of Appendix Table 8 show that high-informality markets experienced a larger increase in the share of firms with 3 or fewer employees and a larger decline in the share of firms with more than 25 employees. These shifts were largest in the 2010s and occurred alongside the widening of firm wage inequality in high-informality markets.

The association between initial informality and the share of salaried workers earning below the minimum wage also strengthened, suggesting weaker enforcement of labor regulations. Column (3) of Appendix Table 8 shows that the association between informality and the share earning below the minimum wage is larger in every period relative to 1995–99, and largest in 2005–09 when the minimum wage increased sharply.

Two checks reduce concerns that estimation bias drives these findings. First, local market fixed effects absorb permanent differences in plug-in bias. A remaining concern is that bias grows more in initially high-informality markets as their firms become smaller. This pattern would bias downward the estimated covariance between worker and firm effects in those markets, making the differential change in the covariance increasingly negative. Instead, column (4) of Appendix Table 9 shows the opposite pattern, with the association becoming more positive over the period.⁴⁶ Second, the results for firm size, the firm size distribution, and earnings dispersion do not rely on estimated firm effects and display the same pattern.

Initially high-informality markets experienced larger increases in firm wage inequality, concentrated in the lower tail, and shifts toward smaller formal firms. These differences were largest during the 2010s stagnation and align with the concentration of the aggregate increase in firm wage inequality among smaller firms documented in Section 4.2. Together with the evidence on minimum-wage compliance, these findings suggest that weak enforcement may have allowed low-paying firms to fall behind as macroeconomic conditions deteriorated.

⁴⁶Appendix Table 9 reports results for the variance of firm effects, worker effects, and their covariance. The point estimates for all three components are positive in the 2010s. Firm effects account for about 39% of the differential increase in the variance of log earnings in the formal sector.

6 Conclusions

This paper examines the role of firms in driving fluctuations in Argentina’s earnings inequality across the 2000s and 2010s, a period of large macroeconomic and institutional changes. During the 2000s expansion, compression in both worker and firm effects contributed to declining inequality. During the 2010s stagnation, rising dispersion in firm wage effects more than accounted for the increase in earnings inequality, despite continued compression in worker effects. This reversal was concentrated among small firms and at the bottom of the firm-effect distribution, suggesting that firm wage setting can make earlier inequality reductions fragile when macroeconomic conditions deteriorate.

Industry differences and labor market institutions help explain these patterns. Widening industry wage differentials account for nearly half of the increase in firm-effect variance during the 2010s, and less productive industries exhibit both lower wage premia and greater dispersion within them. Negotiated wage floors compressed firm wage effects from below during the 2000s, but this compression weakened in the 2010s as the real minimum wage eroded. Local labor markets with higher informality rates subsequently experienced larger increases in dispersion, driven by low-paying firms falling behind. Together, these findings suggest that productivity differences and the effective reach of wage-setting institutions shape how macroeconomic fluctuations translate into earnings inequality.

Several questions remain open. First, extending our Argentina–USA comparison to other developing and developed countries could establish whether industries systematically contribute more to firm wage inequality in developing economies. Second, broader longitudinal data on firm productivity, particularly for small firms, would help assess whether widening productivity dispersion underlies the divergence in firm wage effects and distinguish changes within firms from entry and exit. Finally, combining information on formal and informal employment with direct measures of enforcement would help distinguish weaker compliance with wage floors from productivity changes and employment reallocation.

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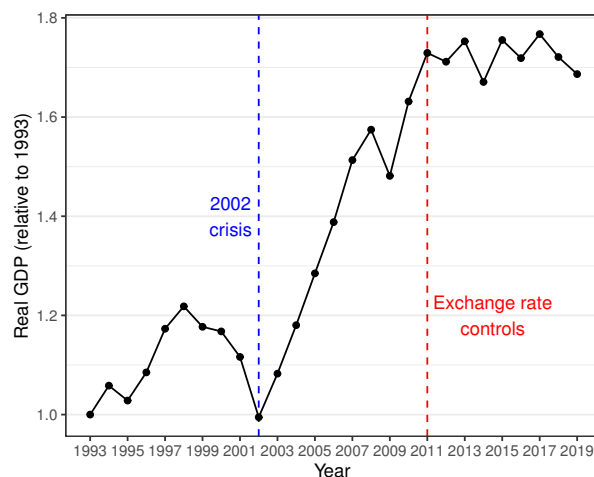
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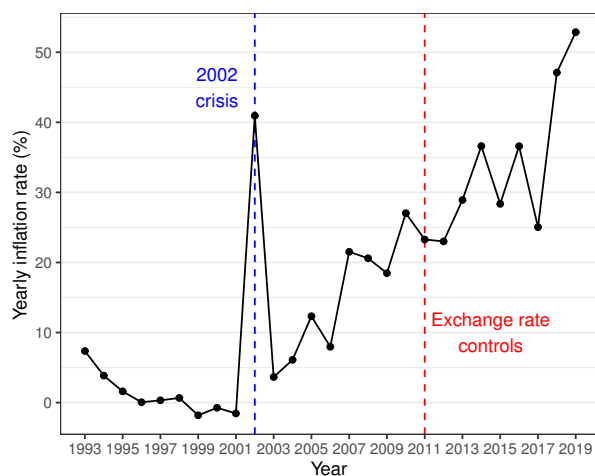
Additional Tables and Figures

Figure 1: Macroeconomic indicators for Argentina, 1993–2019

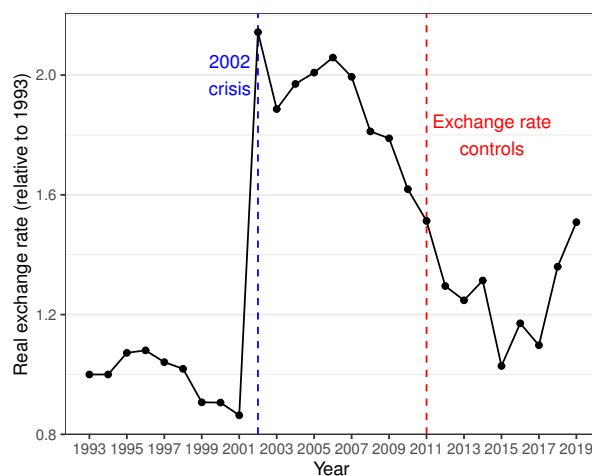
Panel A: Real gross domestic product



Panel B: Yearly inflation

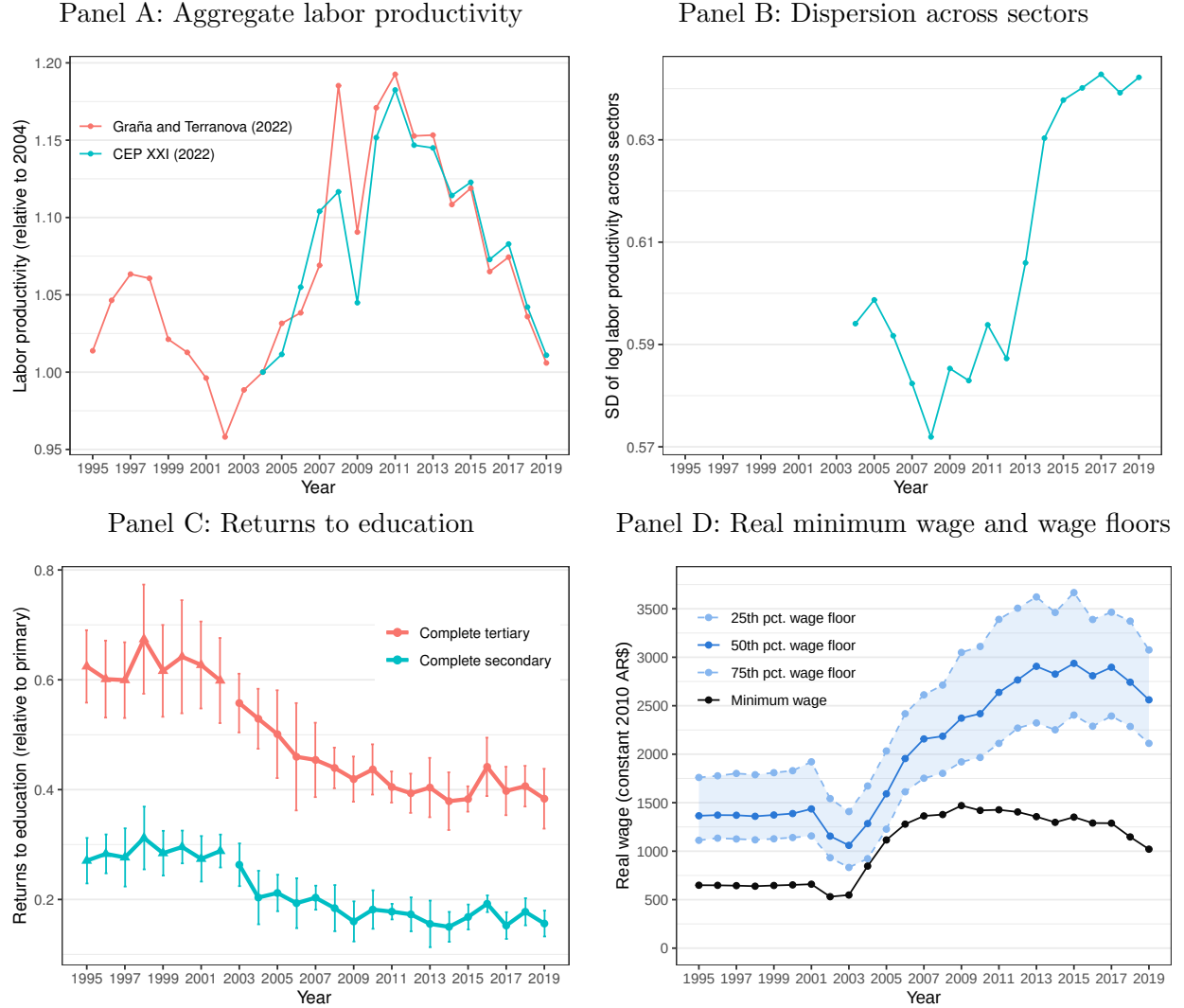


Panel C: Real exchange rate (pesos per USD)



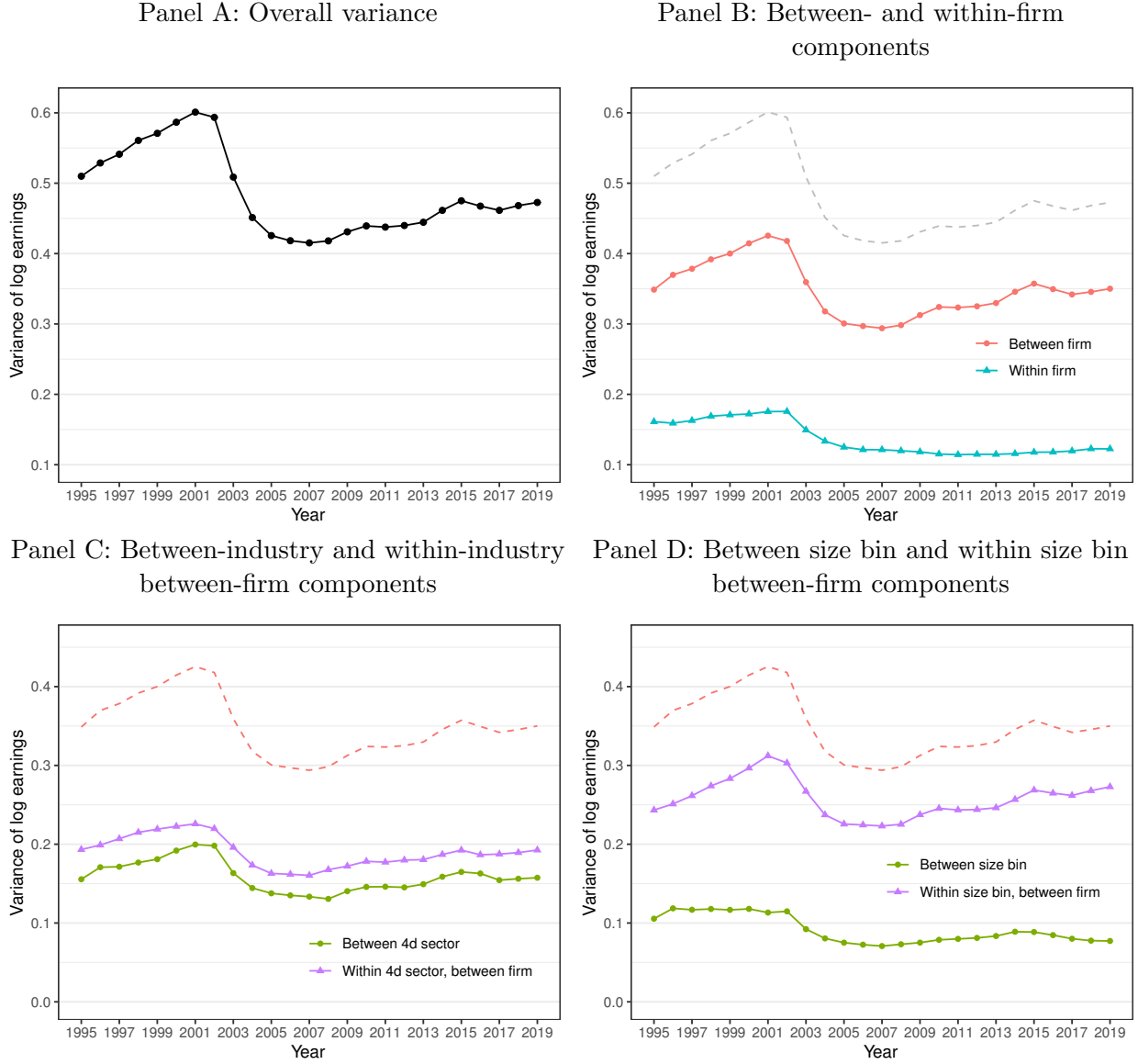
Notes: Data are from the National Institute of Statistics and Censuses (INDEC) and the National Accounts Office (DNCN). The figure shows the evolution of key macroeconomic indicators for Argentina between 1993 and 2019. Panel A shows the evolution of the real Gross Domestic Product (GDP). Panel B shows the yearly inflation rate, constructed as the percent change in the Consumer Price Index (CPI) between December of the previous year and December of the current year. Panel C shows the evolution of the real exchange rate. We use alternative sources for the CPI index between 2007 and 2015 as the official data are widely considered to have been manipulated.

Figure 2: Correlates of wage inequality in Argentina, 1995–2019



Notes: Panel A shows aggregate labor productivity, measured as real output per worker normalized to 2004 = 1. The series from [Graña and Terranova \(2022\)](#) uses real GDP and total employment, covering 1995–2019, while the series from CEP XXI starts in 2004 and aggregates sectoral real value added and employment. Panel B shows the standard deviation of log labor productivity across the 34 sectors of the CEP XXI data in 2004–2019, weighting each sector by its number of jobs. Panel C shows the coefficients of a Mincerian wage regression within a given year controlling for gender, age (linear and quadratic), region, and sector, estimated on data from the Permanent Household Survey (EPH). The omitted category is workers with incomplete secondary education or less (less than 12 years of education); “Complete secondary” corresponds to complete secondary and incomplete tertiary education, and “Complete tertiary” to complete tertiary education. Robust standard errors are clustered at the region level. Different shape markers indicate the change in the EPH survey format in 2003. Panel D shows the real minimum wage, obtained from the Secretary of Labor and deflated by the CPI, and the percentiles 25, 50, and 75 of average wage floors, obtained from resolutions of the Secretary of Labor.

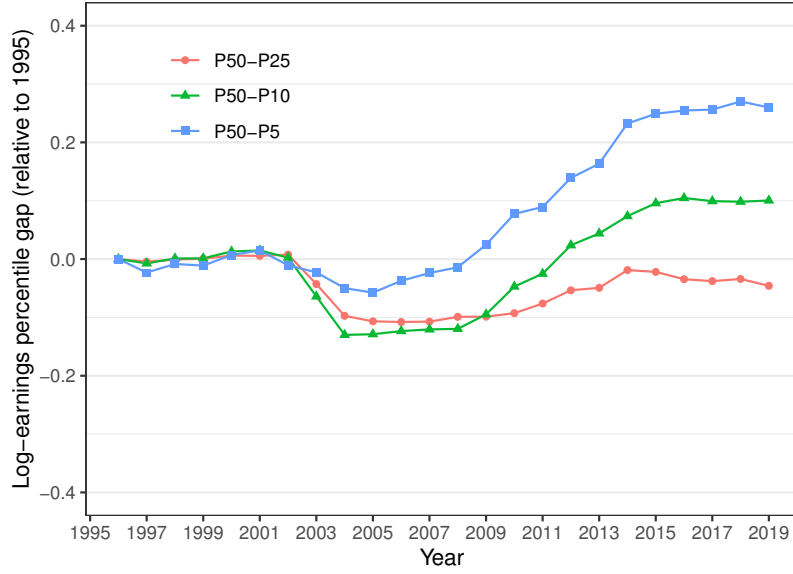
Figure 3: Variance of log earnings in the formal labor market, prime-aged men



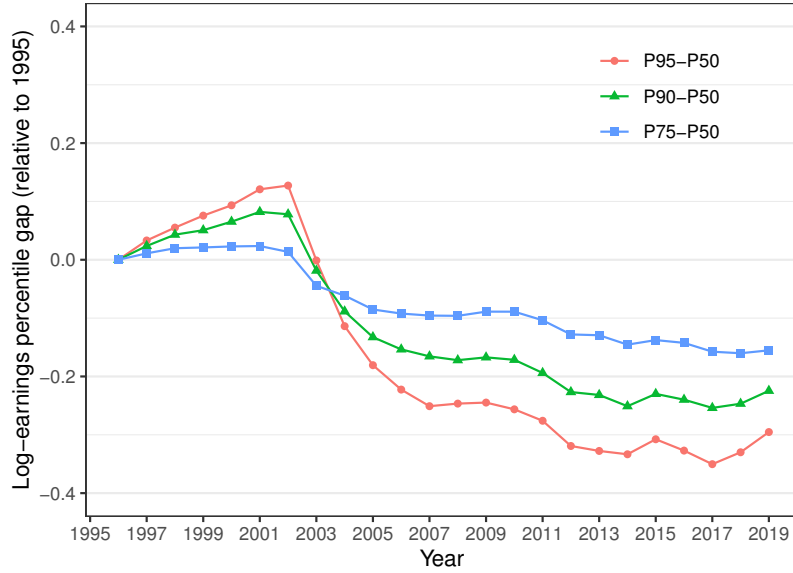
Notes: Data are from Argentina's social security system (SIPA) for individuals aged 20 to 55. The figure shows the evolution of inequality, defined as the variance of log earnings, between 1995 and 2019. Panel A shows the overall variance of log earnings, and Panel B decomposes the variance into the between- and within-firm components. Panel C decomposes the between-firm component into between-industry and within-industry between-firm components. Similarly, Panel D decomposes the between-firm component into between-size-bin and within-size-bin between-firm components. Earnings are measured as average monthly total real earnings. We exclude earnings above the 99.9th percentile of the distribution, and below 75 USD per month, and worker-firm-year cells that correspond to jobs shorter than 6 months and those labeled as internships.

Figure 4: Percentile gaps in log earnings in Argentina relative to 1995, prime-aged men

Panel A: Lower-tail inequality

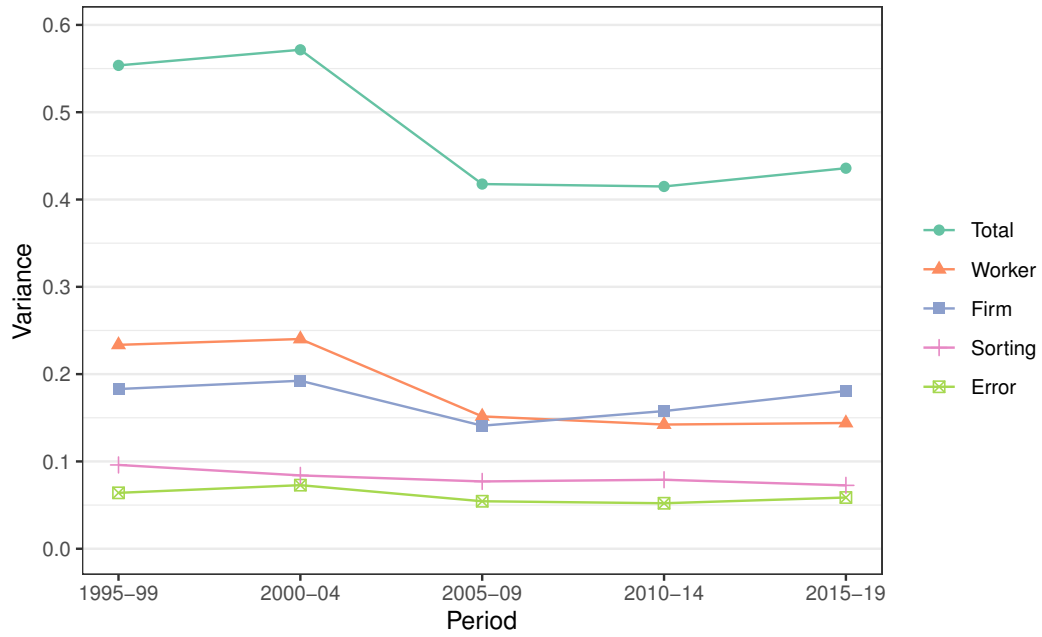


Panel B: Upper-tail inequality



Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure shows the evolution of inequality as measured by differences between percentiles of the log real earnings distribution. Earnings are measured as average monthly total real earnings. We exclude earnings above the 99.9th percentile of the distribution, and below 75 USD per month, and worker-firm-year cells that correspond to jobs lasting fewer than six months or internships.

Figure 5: Variance decomposition of AKM estimates by period, leave-out-match set



Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure visualizes the estimates reported in Table 2 for the variance decomposition of log earnings into worker effects, firm effects, year effects, covariance terms, and the residual.

Figure 6: Firm wage effects and sectoral productivity in 2005–09 and 2015–19

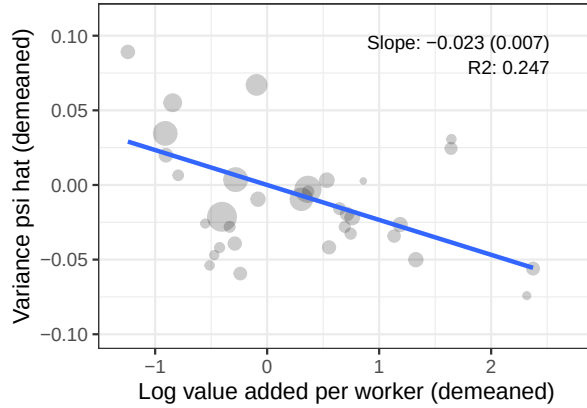
Panel A: Mean firm effect, 2005–09



Panel B: Mean firm effect, 2015–19



Panel C: Variance of firm effects, 2005–09

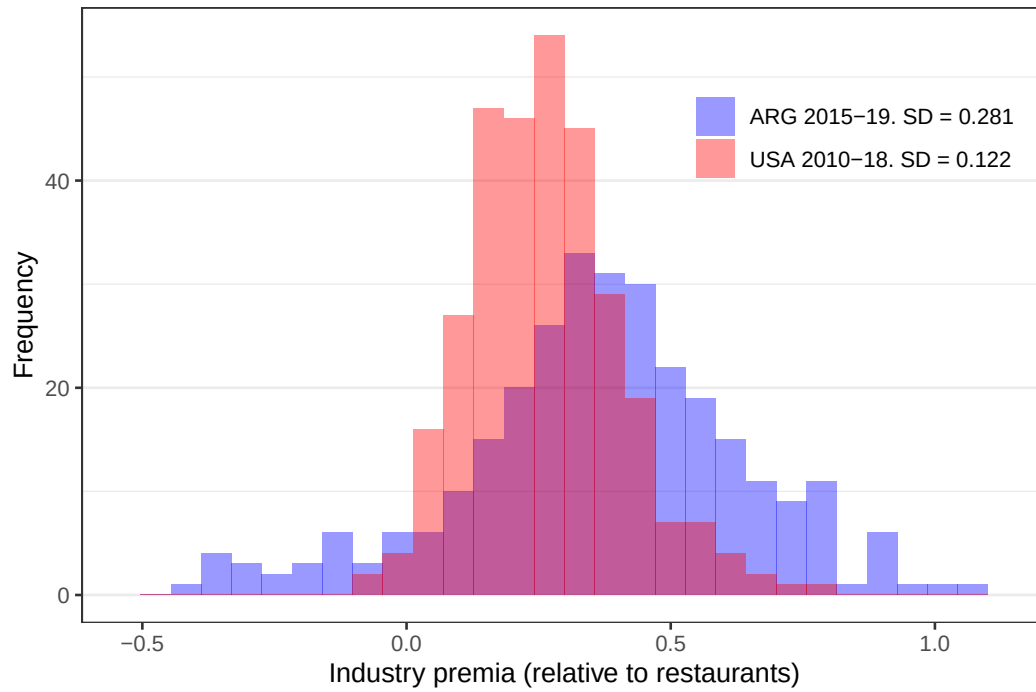


Panel D: Variance of firm effects, 2015–19



Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55 and from Centro de Estudios para la Producción (CEP XXI). The figure shows the relationship between firm effects and sectoral productivity in Argentina for the periods 2005–09 and 2015–19. Panels A and B show a scatter of the average firm wage effect and sectoral productivity, while Panels C and D show a scatter plot of the variance of firm wage effects and sectoral productivity. The measure of sectoral productivity is an average of value added per worker across relevant years. Sectors follow CEP XXI's classification based on ISIC Rev. 3.1, with selected aggregations and subdivisions.

Figure 7: Distribution of industry wage premia, Argentina and the USA



Notes: Data are from Argentina's social security system (SIPA) for 2015–2019 and from [Card et al. \(2024\)](#) for the USA in 2010–2018. The figure shows the distribution of industry wage premia in Argentina and the USA, with industry premia calculated relative to restaurants in both cases. Industries are defined at the 4-digit level in both cases: ISIC Revision 3.1 for Argentina and NAICS for the USA.

Table 1: Summary statistics of SIPA estimation sample, prime-aged men

Period	Millions of			Log earnings		Age		Mean n. firms	Share movers
	Worker years	Workers	Firms	Mean	SD	Mean	SD		
Panel A: Entire sample									
1995–99	10.97	3.51	0.44	7.73	0.73	35.64	9.56	1.273	0.232
2000–04	10.38	3.31	0.42	7.68	0.74	35.99	9.42	1.236	0.205
2005–09	14.10	4.36	0.50	8.00	0.65	35.51	9.38	1.292	0.245
2010–14	15.99	4.74	0.52	8.19	0.66	35.91	9.21	1.273	0.228
2015–19	16.20	4.72	0.50	8.16	0.68	36.83	9.06	1.255	0.213
Panel B: Connected set									
1995–99	8.87	2.36	0.21	7.82	0.74	35.73	9.39	1.391	0.330
2000–04	8.20	2.13	0.18	7.78	0.75	36.01	9.24	1.349	0.301
2005–09	12.00	3.13	0.26	8.06	0.65	35.50	9.20	1.396	0.331
2010–14	13.67	3.44	0.27	8.25	0.65	35.90	9.02	1.366	0.305
2015–19	13.71	3.42	0.23	8.23	0.66	36.79	8.87	1.343	0.285
Panel C: Leave-out connected set									
1995–99	8.08	2.15	0.11	7.87	0.74	35.74	9.38	1.378	0.321
2000–04	7.46	1.94	0.09	7.82	0.76	35.99	9.23	1.335	0.290
2005–09	11.18	2.92	0.15	8.09	0.65	35.45	9.19	1.383	0.321
2010–14	12.75	3.21	0.15	8.28	0.64	35.85	9.01	1.354	0.295
2015–19	12.78	3.19	0.13	8.26	0.66	36.75	8.85	1.333	0.277

Notes: Data are from Argentina’s social security system (SIPA) for men aged 20 to 55. The table shows summary statistics of the administrative data for different periods. Panel A shows the entire sample, Panel B shows the largest connected set of firms, and Panel C shows the largest set of firms that remains connected after we leave out any worker–firm pair. Real earnings are computed as the mean of monthly total real earnings for every worker–firm pair within each year. We then exclude real earnings above the 99.9th percentile of the distribution, and below 75 USD per month, within each year. We also exclude worker–firm–year cells that correspond to jobs shorter than 6 months and those labeled as internships. The latest two columns show the average number of firms per worker and the share of workers with more than one firm in the period.

Table 2: Variance decomposition of AKM estimates by period, leave-out-match estimates

	1995–99		2000–04		2005–09		2010–14		2015–19		Change 1995–2009		Change 2005–2019	
	(1)		(2)		(3)		(4)		(5)		(6)		(7)	
Variance of log earnings	0.554	(100.0)	0.572	(100.0)	0.418	(100.0)	0.415	(100.0)	0.436	(100.0)	-0.136	(100.0)	0.018	(100.0)
Variance of worker effects	0.234	(42.2)	0.240	(42.0)	0.152	(36.3)	0.142	(34.3)	0.144	(33.0)	-0.082	(60.3)	-0.008	(-41.8)
Variance of firm effects	0.183	(33.0)	0.192	(33.7)	0.141	(33.7)	0.158	(38.0)	0.181	(41.5)	-0.042	(30.9)	0.040	(219.0)
Variance of year effects	0.002	(0.3)	0.009	(1.6)	0.016	(3.7)	0.004	(0.9)	0.001	(0.2)	0.014	(-10.1)	-0.015	(-81.1)
2×Cov. of worker and firm effects	0.096	(17.3)	0.084	(14.7)	0.077	(18.5)	0.079	(19.0)	0.073	(16.7)	-0.019	(13.9)	-0.004	(-24.8)
2×Cov. of worker and year effects	-0.002	(-0.4)	-0.001	(-0.2)	-0.007	(-1.6)	-0.002	(-0.6)	0.000	(0.1)	-0.004	(3.2)	0.007	(39.3)
2×Cov. of firm and year effects	-0.001	(-0.3)	-0.001	(-0.2)	0.000	(0.1)	-0.000	(-0.0)	0.000	(0.0)	0.002	(-1.4)	-0.000	(-2.3)
Variance of residual	0.043	(7.8)	0.048	(8.4)	0.039	(9.3)	0.035	(8.3)	0.037	(8.6)	-0.004	(3.2)	-0.001	(-8.2)
Millions of ...														
...worker years	8.08		7.46		11.18		12.75		12.78					
...firms	0.11		0.09		0.15		0.15		0.13					
R^2	0.955		0.949		0.943		0.950		0.951					

Notes: Data are from Argentina’s social security system (SIPA) for men aged 20 to 55. The table shows estimates of the variance decomposition of models of log earnings on worker fixed effects, firm fixed effects, and year fixed effects following [Kline et al. \(2020\)](#). The models are estimated separately for the periods 1995–99, 2000–04, 2005–09, 2010–14, and 2015–19, using the largest leave-out-match connected set of firms in each period. Columns (1) through (5) show the estimates of the variance of each component for each period. Column (6) shows the change in the variance of each component between 1995–99 and 2005–09, and column (7) shows the change between 2005–09 and 2015–19, both in levels and as a percentage of the change in the variance of log earnings. The top eight rows show estimates of the variance of different components of log earnings, with the numbers in parentheses showing the variance normalized by the variance of log earnings in the period or the change in the variance of log earnings between two periods. The bottom three rows show the number of worker years, the number of firms, and the R^2 of the model.

Table 3: Variance of AKM firm wage effects by firm size and period, leave-one-out estimates

	1995–99	2000–04	2005–09	2010–14	2015–19	Change 1995–2009	Change 2005–2019
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Variance of firm effects by size							
Small firms (≤ 25)	0.167	0.177	0.131	0.176	0.215	-0.036 (74.9)	0.085 (190.7)
Medium firms (> 25 to 100)	0.194	0.203	0.143	0.162	0.185	-0.052 (106.9)	0.043 (96.2)
Large firms (> 100)	0.120	0.147	0.104	0.106	0.124	-0.016 (33.2)	0.020 (45.0)
Panel B: Decomposition of total variance							
Total (within+between)	0.199	0.212	0.151	0.171	0.195	-0.048 (100.0)	0.044 (100.0)
Between bin size	0.050	0.045	0.031	0.037	0.040	-0.019 (39.5)	0.009 (19.5)
Within bin size	0.149	0.167	0.120	0.134	0.155	-0.029 (60.5)	0.036 (80.5)
Panel C: Share of worker-years by size							
Small firms (≤ 25)	0.230	0.207	0.234	0.214	0.189	0.004	-0.046
Medium firms (> 25 to 100)	0.241	0.239	0.239	0.228	0.229	-0.002	-0.010
Large firms (> 100)	0.529	0.553	0.527	0.558	0.582	-0.003	0.056

Notes: Data are from Argentina’s social security system (SIPA) for men aged 20 to 55. The table shows the decomposition of the variance of firm wage effects by firm size bin. Firms are classified into three size bins based on average employment: small (25 or fewer workers), medium (more than 25 and at most 100 workers), and large (more than 100 workers). Panel A reports the variance of firm effects estimated separately within each size bin. Panel B decomposes the variance of firm effects into a between-bin component, which captures differences in mean firm effects across size classes, and a within-bin component, which captures dispersion among firms of similar size. The total in Panel B is computed as the sum of the within-bin and between-bin components, so it is internally consistent with the decomposition shown in the table. Because the size bins are estimated separately, this total differs from the economy-wide variance of firm effects reported in Table 2. Numbers in parentheses in columns (6) and (7) express each change as a percentage of the change in the total variance of firm effects in Panel B over the same period. Panel C reports the share of worker-years accounted for by each size bin.

Table 4: Industry wage differentials and the evolution of firm wage effects by period

	1995–99	2005–09	2015–19	Change 1995–2009		Change 2005–2019	
	(1)	(2)	(3)	(4)		(5)	
Panel A: Variance of firm effects, leave-out-match sample							
KSS corrected	0.183	0.141	0.181	-0.042		0.040	
Plug-in	0.195	0.150	0.191	-0.045 (100.0)		0.041	(100.0)
Panel B: Decomposition of plug-in variance							
Between industries	0.075	0.060	0.078	-0.015 (33.0)		0.018	(45.2)
Within industries	0.120	0.090	0.112	-0.030 (67.0)		0.022	(54.8)
Between share (%)	38.4	40.0	41.1				

Notes: Data are from Argentina’s social security system (SIPA) for men aged 20 to 55. Panel A reports the variance of firm wage effects in the largest leave-out-match connected set of firms. The first row uses the heteroskedastic correction of [Kline et al. \(2020\)](#), while the second row reports the corresponding plug-in variance estimate on the same sample. Panel B decomposes the plug-in variance of firm effects into the variance of industry wage differentials, defined as the mean firm effect within each 4-digit industry weighted by worker-years, which corresponds to the between-industries component, and the variance of firm effects within industries. Columns (4) and (5) show changes between 1995–99 and 2005–09, and between 2005–09 and 2015–19. Numbers in parentheses in Panel B express each change as a percentage of the corresponding change in the plug-in variance of firm effects. The last row of Panel B reports the between-industries component as a percentage of the plug-in variance of firm effects in each period. By construction, the between-industries and within-industries components add up to the plug-in variance of firm effects.

Table 5: Negotiated wage floors and changes in earnings and AKM effects, 1995–2009

Panel A: Standard deviation of log earnings, firm effects, and worker effects						
	SD log earnings		SD firm effects		SD worker effects	
	OLS	IV	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Change in mean	-0.0754	-0.0956	-0.0805	-0.1269	-0.0407	-0.0583
log wage floor	(0.0311)	(0.0467)	(0.0436)	(0.0665)	(0.0238)	(0.0378)
R^2	0.0973	0.0886	0.1013	0.1427	0.0763	0.0886
Observations	43	43	43	43	43	43
Panel B: Percentiles of firm-effect distribution						
	10th percentile		25th percentile		90th percentile	
	OLS	IV	OLS	IV	OLS	IV
	(7)	(8)	(9)	(10)	(11)	(12)
Change in mean	0.2540	0.4127	0.1971	0.3783	0.0223	0.0116
log wage floor	(0.0941)	(0.1414)	(0.1050)	(0.1354)	(0.0621)	(0.0825)
R^2	0.1836	0.2749	0.1300	0.2715	0.0033	0.0005
Observations	43	43	43	43	43	43

Notes: Data are from Argentina’s social security system (SIPA) for men aged 20 to 55 and from Secretary of Labor resolutions reporting average wage floors. The table shows estimates of the effect of changes in negotiated wage floors on changes in inequality measures at the ISIC Revision 3.1 two-digit industry level between 1995–99 and 2005–09. The industry’s mean log negotiated wage floor is computed between 2001–02 and 2008–09, and uses bargaining units observed consistently from the late 1990s through 2008 and sectors with at least two wage-floor observations. The dependent variables are changes between 1995–99 and 2005–09 in the standard deviation (SD) of log earnings, the SD of firm effects, the SD of worker effects, and the 10th, 25th, and 90th percentiles of the firm-effect distribution. Odd-numbered columns report OLS estimates. Even-numbered columns report IV estimates that instrument the change in the mean log wage floor with the negative gap between the industry’s mean log wage floor in 2001–02 and the average log real national minimum wage in 2005–07. Heteroskedasticity-robust standard errors are reported in parentheses.

Table 6: Informality, the dispersion of firm wage effects, and firm size

	SD log earnings (1)	SD firm effects (2)	10th ptile. firm effect (3)	90th ptile. firm effect (4)	Log mean firm size (5)
Informality $\times$ 2000–04	0.004 (0.019)	-0.005 (0.021)	-0.069 (0.042)	-0.135 (0.074)	-0.138 (0.134)
Informality $\times$ 2005–09	0.122 (0.045)	0.079 (0.037)	-0.144 (0.079)	-0.031 (0.103)	-0.206 (0.146)
Informality $\times$ 2010–14	0.194 (0.059)	0.135 (0.052)	-0.271 (0.069)	-0.005 (0.122)	-0.434 (0.164)
Informality $\times$ 2015–19	0.193 (0.062)	0.121 (0.058)	-0.304 (0.099)	-0.053 (0.104)	-0.432 (0.186)
Time and local market effects	Yes	Yes	Yes	Yes	Yes
R^2	0.877	0.800	0.941	0.907	0.974
Observations	335	335	335	335	335

Notes: Data are from Argentina’s social security system (SIPA) for men aged 20 to 55 and from the Permanent Household Survey (EPH). The table shows estimates of the association between labor informality and the dispersion of earnings and firm wage effects across local labor markets, together with average firm size. The unit of observation is a region-sector cell, defined by the seven EPH regions and the sections of ISIC Revision 3.1, some of which are combined to match the EPH classification. The dependent variables are the standard deviation of log earnings in column (1), the standard deviation and the 10th and 90th percentiles of the estimated AKM firm effects in columns (2) through (4), and the log of average firm size in column (5). The regressor is the informality rate in 1995–99, defined as the share of salaried workers not registered with the social security system, interacted with period indicators. The 1995–99 interaction is omitted, so the table reports differential changes relative to that period. The estimation sample is restricted to cells with at least 30 EPH observations. Firm effects are measured with the plug-in estimator, and firm size is computed over the connected set of men. All specifications include period and region-sector fixed effects, and observations are weighted by the number of worker-years in the cell. Standard errors clustered at the region-sector level are reported in parentheses.

Appendix for

“Firms and the Evolution of Earnings Inequality in Argentina’s Volatile Economy”

(not for publication)

A Data Appendix

This appendix provides additional details on data sources, variable construction, and sample restrictions.

A.1 Administrative Data (SIPA)

The *Sistema Integrado Previsional Argentino* (SIPA) contains administrative records of all employment relationships registered with Argentina’s social security system. The data cover the period 1995–2019 and include all private-sector wage employment, employment in state-owned enterprises (e.g., YPF, Aerolíneas Argentinas), and employment in autonomous public institutions (e.g., national universities, provincial banks). The data exclude public-sector employees in central, provincial, and municipal government, as well as self-employed workers and informal employees. Each record contains a unique worker identifier and firm identifier that allow us to track individuals across employers over time.

Earnings Variable. For each worker-firm-month observation, we observe total pre-tax compensation, which includes base salary, regular bonuses (e.g., the mandatory 13th-month salary), overtime pay, severance payments (when applicable), and other taxable compensation.

We process earnings as follows. First, we convert nominal earnings to constant December 2010 pesos using the Consumer Price Index (CPI), accounting for the manipulation of official statistics in 2007–2015. Second, we construct an average monthly earnings variable for each worker spell. Third, we exclude observations above the 99.9th percentile of the earnings distribution within each year to limit the influence of outliers, such as unusually large severance payments or bonuses. Fourth, we exclude observations with monthly earnings below the equivalent of \$75 USD at 2010 prices (approximately 318 Argentine pesos at 2010 prices), a

threshold roughly one-third of the minimum wage intended to exclude workers with minimal labor force attachment. We verify robustness to alternative thresholds (\$100, \$125, \$150, \$200 USD) in the appendix.⁴⁷

Sample Restrictions. We restrict the sample to men aged 20 to 55, though in the appendix we show results for women. We exclude worker-firm-year observations corresponding to jobs that lasted fewer than 6 months within the calendar year, as well as observations explicitly coded as internships or training contracts using the hiring modality variable (*modalidad de contratación*) of the SIPA data. We also exclude workers employed by temporary employment agencies, who appear in the data as employed by both the agency and the client firm, and firms classified as temporary employment agencies in the BADE (Base de Análisis de la Dinámica del Empleo) firm registry. Employment in state-owned enterprises and autonomous public institutions is retained. When a worker holds multiple jobs within a year, we keep only the highest-paying job, following standard practice in the AKM literature.

Firm and Worker Characteristics. The administrative data include firm-level information on 4-digit industry classification (based on ISIC Rev. 3.1) and the province where the firm is registered (available from the BADE firm registry). At the worker level, the data contain age (computed from year of birth recorded in the social security system) and gender. Worker education is not available.

Connected Sets. Estimation of the AKM models requires a set of firms connected by worker mobility. We define two connected sets. The first is the largest connected set: the largest set of firms such that any two firms can be linked through a chain of worker moves. The second is the leave-one-out connected set: the largest set of firms that remains connected after removing any single worker-firm match, a more restrictive sample required for the bias correction method of [Kline et al. \(2020\)](#). Following standard practice, we drop single stayers—worker-firm pairs observed only once—before computing connected sets, as these observations do not contribute to identification of either worker or firm effects.

A.2 Household Survey Data (EPH)

The *Encuesta Permanente de Hogares* (EPH) is Argentina’s main labor force survey, conducted by the National Institute of Statistics and Censuses (INDEC). The survey underwent a major methodological change in 2003: the original format (“Puntual” EPH, 1995–2002)

⁴⁷A useful reference is the 1990s national minimum wage, which was fixed at 200 nominal ARS, equivalent to \$200 USD.

was conducted twice per year (May and October) in 28 urban agglomerations, while the revised format (“Continua” EPH, 2003–2019) collects data quarterly in 31 urban agglomerations. We use the October wave for 1995–2002 and the fourth quarter for 2003–2019, except for 2015 when we use the second quarter due to data availability limitations (the survey was temporarily suspended).

Variable Definitions. We classify individuals as employed if they report having worked at least one hour in the reference week or having a job from which they were temporarily absent.⁴⁸ We define informal salaried employment as jobs where the employer does not make regular contributions to the social security system. This definition excludes self-employment, which accounts for approximately one-quarter of the labor force. Earnings correspond to monthly after-tax labor income from the main job (*ingreso de la ocupación principal*), and are also deflated to 2010 ARS using the CPI. To classify workers as earning below the minimum wage, we compare their hourly earnings with the hourly national minimum wage, defined as the monthly minimum wage published by the Secretary of Labor divided by 200 hours.⁴⁹ Establishment size is reported by workers as the number of employees at their workplace, which we group into four categories: 1–5, 6–25, 26–100, and more than 100 employees.

We construct six education categories based on the highest level attained: primary incomplete (less than 7 years of education), primary complete (7 years), secondary incomplete (8–11 years), secondary complete (12 years), tertiary incomplete (13–16 years), and tertiary complete (17+ years). Industry codes, which vary across survey waves, are harmonized to a consistent classification at the section level of ISIC Rev. 3.1.

Sample Restrictions. For comparability with the administrative data, we restrict the EPH sample to salaried workers (excluding self-employed and employers) aged 20 to 55 with positive reported earnings and valid responses on formality status.

A.3 Firm Survey Data (ENDEI)

The first wave of the *Encuesta Nacional de Dinámica de Empleo e Innovación* (ENDEI), conducted in 2012, collects data on firm activities during 2010–12. The survey is representa-

⁴⁸For completeness, we note that individuals are classified as unemployed if they are not employed and are actively seeking work in the reference week. We show the evolution of the employment-to-population ratio and the unemployment rate in Appendix Figure 1.

⁴⁹The “Puntual” EPH reports hourly earnings directly, which we compare with the minimum wage in force in the fourth quarter of each year. For the “Continua” EPH we divide monthly earnings in the main job by usual weekly hours times 30/7, and compare them with the quarterly average of the minimum wage.

tive of manufacturing firms with 10 or more employees, covering approximately 3,700 firms in the first round. Key variables include value added (total sales minus intermediate inputs, reported annually) and employment (number of employees), which allow computation of value added per worker as a productivity measure. Firm identifiers allow us to match these firms with the SIPA data.

B Additional Tables and Figures

Appendix Table 1: Summary statistics of SIPA estimation sample, prime-aged women

Period	Millions of			Log earnings		Age		Mean n. firms	Share movers
	Worker years	Workers	Firms	Mean	SD	Mean	SD		
Panel A: Entire sample									
1995–99	4.27	1.44	0.26	7.56	0.65	34.55	9.66	1.201	0.177
2000–04	4.37	1.42	0.24	7.54	0.67	35.07	9.42	1.184	0.164
2005–09	5.63	1.85	0.30	7.83	0.60	34.88	9.24	1.214	0.185
2010–14	6.69	2.11	0.33	8.01	0.61	35.48	9.04	1.198	0.172
2015–19	6.95	2.12	0.33	8.02	0.63	36.72	8.87	1.183	0.160
Panel B: Connected set									
1995–99	3.02	0.83	0.08	7.67	0.67	34.53	9.48	1.315	0.274
2000–04	3.14	0.81	0.07	7.64	0.69	34.96	9.22	1.290	0.256
2005–09	4.38	1.17	0.11	7.90	0.60	34.80	9.02	1.319	0.275
2010–14	5.23	1.34	0.12	8.09	0.60	35.42	8.83	1.291	0.251
2015–19	5.36	1.36	0.10	8.11	0.62	36.63	8.66	1.266	0.231
Panel C: Leave-out connected set									
1995–99	2.65	0.73	0.04	7.71	0.69	34.57	9.48	1.291	0.255
2000–04	2.77	0.71	0.03	7.67	0.70	34.93	9.21	1.272	0.240
2005–09	3.95	1.06	0.05	7.92	0.61	34.77	9.00	1.297	0.257
2010–14	4.73	1.22	0.06	8.11	0.61	35.40	8.81	1.269	0.233
2015–19	4.85	1.23	0.05	8.14	0.62	36.60	8.63	1.248	0.216

Notes: Data are from Argentina’s social security system (SIPA) for women aged 20 to 55. The table shows summary statistics of the administrative data for different periods. Panel A shows the entire sample, Panel B shows the largest connected set of firms, and Panel C shows the largest set of firms that remains connected after we leave out any worker–firm pair. Real earnings are computed as the mean of monthly total real earnings for every worker–firm pair within each year. We then exclude real earnings above the 99.9th percentile of the distribution, and below 75 USD per month, within each year. We also exclude worker–firm–year cells that correspond to jobs shorter than 6 months and those labeled as internships. The latest two columns show the average number of firms per worker and the share of workers with more than one firm in the period.

Appendix Table 2: Variance decomposition of AKM estimates by period, connected set

	1995–99		2000–04		2005–09		2010–14		2015–19		Change 1995–2009		Change 2005–2019	
	(1)		(2)		(3)		(4)		(5)		(6)		(7)	
Variance of log earnings	0.548	(100.0)	0.564	(100.0)	0.416	(100.0)	0.417	(100.0)	0.438	(100.0)	-0.132	(100.0)	0.022	(100.0)
Variance of worker effects	0.266	(48.5)	0.278	(49.3)	0.179	(43.1)	0.172	(41.1)	0.177	(40.4)	-0.087	(65.6)	-0.003	(-11.4)
Variance of firm effects	0.212	(38.8)	0.225	(39.9)	0.163	(39.1)	0.184	(44.0)	0.208	(47.6)	-0.050	(37.6)	0.046	(207.0)
Variance of year effects	0.002	(0.3)	0.009	(1.6)	0.015	(3.7)	0.004	(0.9)	0.001	(0.2)	0.014	(-10.4)	-0.015	(-66.0)
2×Cov. of worker and firm effects	0.048	(8.8)	0.026	(4.7)	0.042	(10.0)	0.040	(9.7)	0.031	(7.0)	-0.006	(4.8)	-0.011	(-49.6)
2×Cov. of worker and year effects	-0.002	(-0.4)	-0.001	(-0.2)	-0.007	(-1.6)	-0.002	(-0.5)	0.000	(0.1)	-0.004	(3.2)	0.007	(31.1)
2×Cov. of firm and year effects	-0.001	(-0.3)	-0.001	(-0.2)	0.000	(0.1)	-0.000	(-0.0)	0.000	(0.0)	0.002	(-1.4)	-0.000	(-1.7)
Variance of residual	0.024	(4.3)	0.028	(4.9)	0.023	(5.5)	0.020	(4.8)	0.021	(4.8)	-0.001	(0.5)	-0.002	(-9.5)
Millions of ...														
...worker years	8.87		8.20		12.00		13.67		13.71					
...firms	0.21		0.18		0.26		0.27		0.23					
R^2	0.957		0.951		0.945		0.952		0.952					

Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The table shows estimates of the variance decomposition of models of log earnings on worker fixed effects, firm fixed effects, and year fixed effects following [Abowd et al. \(1999\)](#). The models are estimated separately for the periods 1995–99, 2000–04, 2005–09, 2010–14, and 2015–19, using the largest connected set of firms in each period. Columns (1) through (5) show the estimates of the variance of each component for each period. Column (6) shows the change in the variance of each component between 1995–99 and 2005–09, and column (7) shows the change between 2005–09 and 2015–19, both in levels and as a percentage of the change in the variance of log earnings. The top eight rows show estimates of the variance of different components of log earnings, with the numbers in parentheses showing the variance normalized by the variance of log earnings in the period or the change in the variance of log earnings between two periods. The bottom three rows show the number of worker years, the number of firms, and the R^2 of the model.

Appendix Table 3: Regressions of firm effects on productivity and size, 2010–12 ENDEI sample

	Dependent variable: firm wage effect								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log VA per worker	0.1446 (0.0166)		0.1028 (0.0062)	0.1317 (0.0211)		0.0954 (0.0137)	0.0984 (0.0158)		0.1190 (0.0092)
Log firm size		0.1131 (0.0088)	0.1030 (0.0233)		0.1089 (0.0093)	0.1060 (0.0084)		0.1033 (0.0080)	0.1946 (0.0139)
Log VA pw * Log firm size			-0.0001 (0.0015)			-0.0006 (0.0006)			-0.0081 (0.0006)
Province FE	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
2-digit sector FE	No	No	No	No	No	No	Yes	Yes	Yes
Num. of firms	3,028	3,028	3,028	3,028	3,028	3,028	3,028	3,028	3,028
Num. of worker-years	1,363,444	1,363,444	1,363,444	1,363,444	1,363,444	1,363,444	1,363,444	1,363,444	1,363,444
R^2	0.1617	0.3725	0.4495	0.2433	0.4496	0.5094	0.3689	0.5424	0.5733
R^2 within				0.1451	0.3782	0.4457	0.0890	0.3394	0.3841

Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55 and from the *Encuesta Nacional de Dinámica de Empleo e Innovación* (ENDEI). The table shows regressions of estimated firm wage effects on log value added per worker, log firm size, and their interaction, for firms observed in the 2010–12 ENDEI sample. Firm effects are obtained from an AKM model estimated on SIPA data for 2010–14. The unit of observation is a worker-year, so that firms are weighted by their number of worker-years, and standard errors are clustered at the two-digit sector and province levels. Value added per worker is calculated from self-reported ENDEI data for 2010–12, averaging the values for those three years to obtain one value for the period. Data below the 1st percentile and above the 99th percentile of the log value-added-per-worker distribution are trimmed to prevent extreme values from driving the results.

Appendix Table 4: Top and bottom 10 industry wage differentials, 2015–19

Rank	Code	Industry name	Mean log earnings	Industry premium	Mean Worker Effect	Percent of workforce
		(1)	(2)	(3)	(4)	(5)
		Across industry mean	8.261	0.390	0.000	
		Across industry SD	0.393	0.281	0.151	
1	1320	Mining of non-ferrous metal ores	9.160	1.129	0.159	0.34
2	3410	Manufacture of motor vehicles	9.030	1.108	0.051	0.93
3	1120	Service activities incidental to oil and gas	9.324	1.072	0.382	1.44
4	1110	Crude oil and gas extraction	9.448	1.007	0.573	0.49
5	6030	Transport via pipelines	9.230	0.956	0.405	0.05
6	4011	Production of electricity	9.263	0.926	0.466	0.30
7	6521	Services of banking financial institutions	9.069	0.917	0.281	1.15
8	2330	Processing of nuclear fuel	8.862	0.913	0.079	0.02
9	2320	Manufacture of refined petroleum products	9.038	0.899	0.265	0.17
10	2230	Reproduction of recorded media	9.082	0.890	0.322	0.00
291	0112	Vegetable growing and nursery	7.448	-0.258	-0.166	0.03
292	4549	Building completion	7.320	-0.268	-0.282	0.01
293	0150	Hunting and game services	7.326	-0.277	-0.268	0.24
294	4542	Finishing and covering of walls and floors	7.261	-0.321	-0.290	0.03
295	0502	Fish farms and other aquatic products	7.651	-0.324	0.104	0.00
296	0503	Fishing services	7.413	-0.337	-0.128	0.00
297	8000	Education	7.288	-0.374	-0.210	0.02
298	4511	Building demolition	7.570	-0.376	0.076	0.00
299	4544	Painting and decorating work	7.630	-0.376	0.135	1.94
300	5242	Retail sale of used books	7.203	-0.430	-0.238	0.05

Notes: Data are from Argentina’s social security system (SIPA), with industry wage differentials calculated following [Card et al. \(2024\)](#). Worker effects are obtained from the AKM model estimated on men aged 20 to 55 for the 2015–19 period. The table shows the 10 industries with the largest and smallest industry wage differentials in 2015–19, at the 4-digit level of disaggregation. Column (1) shows the name of the industry, shortened for display purposes. Column (2) shows the average earnings, column (3) the industry wage differentials, and column (4) the average worker effect, all weighted by the number of worker-years. Column (5) shows the percentage of the total number of worker-years in the given industry, with “0.00” indicating that the industry represents fewer than .01% of worker-years.

Appendix Table 5: Industry wage differentials across broad industries, relative to restaurants

Code	Industry description	Industry wage differential			Share of worker-years		
		1995–99	2005–09	2015–19	1995–99	2005–09	2015–19
A	Agriculture, hunting and forestry	-0.070	-0.085	0.002	0.044	0.054	0.046
B	Fishing	0.649	0.632	0.484	0.003	0.004	0.003
C	Mining and quarrying	0.727	0.904	0.988	0.017	0.021	0.026
D	Manufacturing	0.361	0.367	0.483	0.361	0.289	0.268
E	Electricity, gas and water supply	0.646	0.602	0.738	0.021	0.014	0.018
F	Construction	-0.009	0.127	0.068	0.056	0.099	0.108
G	Wholesale and retail trade	0.228	0.238	0.422	0.133	0.152	0.155
H	Hotels and restaurants	0.084	0.095	0.143	0.021	0.024	0.022
I	Transport, storage and communications	0.517	0.477	0.568	0.136	0.120	0.129
J	Financial intermediation	0.598	0.520	0.735	0.039	0.025	0.023
K	Real estate and business activities	0.154	0.165	0.301	0.085	0.122	0.118
M	Education	-0.518	-0.412	-0.376	0.017	0.017	0.019
N	Health and social work	0.300	0.265	0.406	0.021	0.017	0.023
O	Other community and social service activities	0.271	0.283	0.408	0.045	0.041	0.041

Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The table shows industry wage differentials aggregated at the section level for 1995–99, 2005–09, and 2015–19. Industry wage differentials are computed relative to restaurants. Sections are defined according to ISIC Revision 3.1. The last three columns report each section's share of worker-years in the corresponding period.

Appendix Table 6: Negotiated wage floors and changes in earnings and AKM effects, 1995–2019

Panel A: Standard deviation of log earnings, firm effects, and worker effects						
	SD log earnings		SD firm effects		SD worker effects	
	OLS	IV	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Change in mean	-0.0050	0.0680	-0.0602	-0.0544	-0.0105	0.0437
log wage floor	(0.0432)	(0.0457)	(0.0539)	(0.0731)	(0.0395)	(0.0584)
R^2	0.0003	0.0284	0.0388	0.0179	0.0021	0.0210
Observations	43	43	43	43	43	43
Panel B: Percentiles of firm-effect distribution						
	10th percentile		25th percentile		90th percentile	
	OLS	IV	OLS	IV	OLS	IV
	(7)	(8)	(9)	(10)	(11)	(12)
Change in mean	0.1371	0.1935	0.0993	0.2271	-0.1075	-0.0914
log wage floor	(0.1040)	(0.1298)	(0.0862)	(0.1237)	(0.1027)	(0.1490)
R^2	0.0471	0.0532	0.0346	0.1026	0.0378	0.0155
Observations	43	43	43	43	43	43

Notes: Data are from Argentina’s social security system (SIPA) for men aged 20 to 55 and from Secretary of Labor resolutions reporting average wage floors. The table shows estimates of the effect of changes in negotiated wage floors on changes in inequality measures at the ISIC Revision 3.1 two-digit industry level between 1995–99 and 2015–19. The industry’s mean log negotiated wage floor is computed between 2001–02 and 2008–09, and uses bargaining units observed consistently from the late 1990s through 2008 and sectors with at least two wage-floor observations. The dependent variables are changes between 1995–99 and 2015–19 in the standard deviation (SD) of log earnings, the SD of firm effects, the SD of worker effects, and the 10th, 25th, and 90th percentiles of the firm-effect distribution. Odd-numbered columns report OLS estimates. Even-numbered columns report IV estimates that instrument the change in the mean log wage floor with the negative gap between the industry’s mean log wage floor in 2001–02 and the average log real national minimum wage in 2005–07. Heteroskedasticity-robust standard errors are reported in parentheses.

Appendix Table 7: First-stage and reduced-form estimates for negotiated wage floors

Panel A: First-stage regressions							
	Change in log negotiated wage floor						
	Mean (1)	10th ptile. (2)	25th ptile. (3)	50th ptile. (4)	75th ptile. (5)	90th ptile. (6)	Std. dev. (7)
Negative pre-treatment wage-floor gap	0.5497 (0.0768)	0.7247 (0.0728)	0.6323 (0.0786)	0.5682 (0.0829)	0.4993 (0.0968)	0.3263 (0.1019)	-0.1703 (0.0409)
R^2	0.5671	0.6070	0.6024	0.5668	0.4477	0.1877	0.2356
First-stage F statistic	51.27	99.01	64.76	46.97	26.58	10.25	17.37
Observations	43	43	43	43	43	43	43
Panel B: Initial wage-floor gap and changes in the variance of firm effects							
	1995–99 (8)	2000–04 (9)	2005–09 (10)	2010–14 (11)	2015–19 (12)		
Negative pre-treatment wage-floor gap	0	-0.0213 (0.0143)	-0.0469 (0.0211)	-0.0278 (0.0274)	-0.0180 (0.0266)		
R^2	—	0.0695	0.1323	0.0346	0.0121		
Observations	—	43	43	43	43		

Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55 and from Secretary of Labor resolutions reporting average wage floors. The table shows first-stage and reduced-form estimates of the effect of the wage-floor gap on changes in negotiated wage floors (Panel A) and in the variance of firm effects (Panel B), at the ISIC Revision 3.1 two-digit industry level. The wage-floor gap is the negative gap between the industry's mean log wage floor in 2001–02 and the average log real national minimum wage in 2005–07. In Panel A, the dependent variables are changes between 2001–02 and 2008–09 in the mean, indicated percentiles, and standard deviation of log wage floors, which use bargaining units observed consistently from the late 1990s through 2008 and sectors with at least two wage-floor observations. The first-stage F statistic tests whether the coefficient on the wage-floor gap is zero. In Panel B, the dependent variable is the change in the variance of firm effects between 1995–99 and each indicated period, with 1995–99 normalized to zero. Heteroskedasticity-robust standard errors are reported in parentheses.

Appendix Table 8: Informality, enforcement, and the size distribution of firms

	Share firms ≤ 3 employees (1)	Share firms > 25 employees (2)	Share less than MW (3)
Informality $\times$ 2000–04	0.011 (0.021)	-0.024 (0.006)	0.213 (0.053)
Informality $\times$ 2005–09	0.022 (0.025)	-0.017 (0.009)	0.550 (0.119)
Informality $\times$ 2010–14	0.053 (0.028)	-0.046 (0.010)	0.492 (0.075)
Informality $\times$ 2015–19	0.074 (0.032)	-0.063 (0.012)	0.328 (0.061)
Time and local market effects	Yes	Yes	Yes
R^2	0.979	0.980	0.880
Observations	335	335	322

Notes: Data are from Argentina’s social security system (SIPA) for men aged 20 to 55 and from the Permanent Household Survey (EPH). The table shows estimates of the association between labor informality, compliance with the minimum wage, and the size distribution of firms across local labor markets. The unit of observation is a region-sector cell, defined by the seven EPH regions and the sections of ISIC Revision 3.1, some of which are combined to match the EPH classification. The dependent variables are the shares of firms in the two tails of the size distribution, measured in the social security records, in columns (1) and (2), and the share of salaried workers earning less than the minimum wage, measured in the EPH, in column (3). The regressor is the informality rate in 1995–99, defined as the share of salaried workers not registered with the social security system, interacted with period indicators. The 1995–99 interaction is omitted, so the table reports differential changes relative to that period. The estimation sample is restricted to cells with at least 30 EPH observations. The size distribution is computed over the connected set of men. All specifications include period and region-sector fixed effects, and observations are weighted by the number of worker-years in the cell. Standard errors clustered at the region-sector level are reported in parentheses.

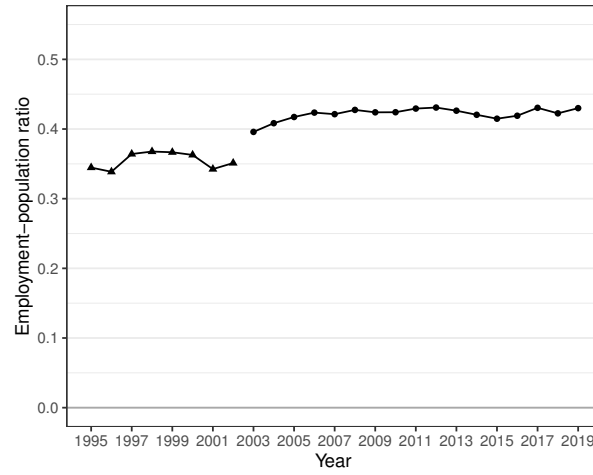
Appendix Table 9: Informality and the variance decomposition of log earnings

	Var. log earnings (1)	Var. firm effects (2)	Var. worker effects (3)	2× Cov. firm, worker effects (4)
Informality × 2000–04	-0.003 (0.026)	-0.005 (0.017)	0.001 (0.013)	-0.000 (0.024)
Informality × 2005–09	0.176 (0.062)	0.061 (0.030)	0.080 (0.016)	0.026 (0.029)
Informality × 2010–14	0.260 (0.082)	0.107 (0.040)	0.109 (0.022)	0.037 (0.036)
Informality × 2015–19	0.260 (0.085)	0.102 (0.045)	0.115 (0.023)	0.040 (0.033)
Time and local market effects	Yes	Yes	Yes	Yes
R^2	0.867	0.798	0.954	0.896
Observations	335	335	335	335

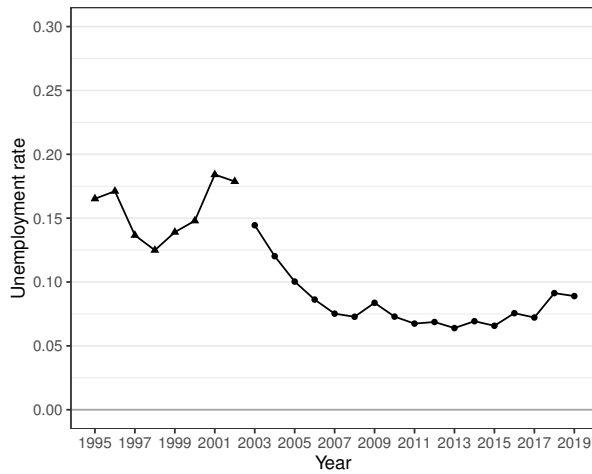
Notes: Data are from Argentina’s social security system (SIPA) for men aged 20 to 55 and from the Permanent Household Survey (EPH). The table shows estimates of the association between labor informality and the variance of log earnings across local labor markets, decomposed into the contributions of worker effects, firm effects, and sorting. The unit of observation is a region-sector cell, defined by the seven EPH regions and the sections of ISIC Revision 3.1, some of which are combined to match the EPH classification. The dependent variables are the variance of log earnings and the components of its decomposition into the variance of the estimated AKM worker and firm effects and twice their covariance. The regressor is the informality rate in 1995–99, defined as the share of salaried workers not registered with the social security system, interacted with period indicators. The 1995–99 interaction is omitted, so the table reports differential changes relative to that period. The estimation sample is restricted to cells with at least 30 EPH observations. Column (1) is the sum of columns (2), (3), and (4), and of the variance of the residual and of the time-varying covariates, so the first three components need not add up to the total. Variances are computed with the plug-in estimator. All specifications include period and region-sector fixed effects, and observations are weighted by the number of worker-years in the cell. Standard errors clustered at the region-sector level are reported in parentheses.

Appendix Figure 1: Labor market indicators for Argentina, 1995-2019

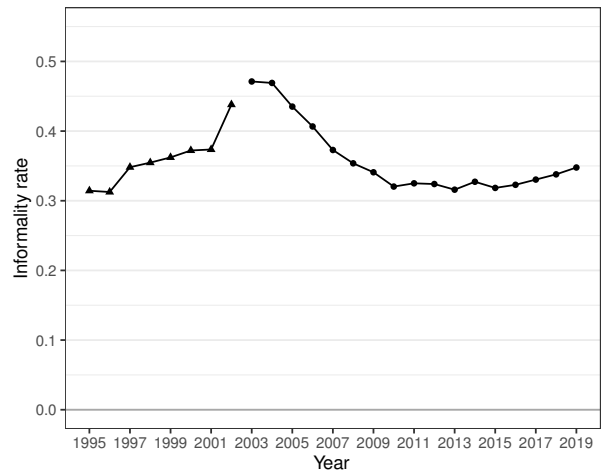
Panel A: Employment to population ratio



Panel B: Unemployment rate

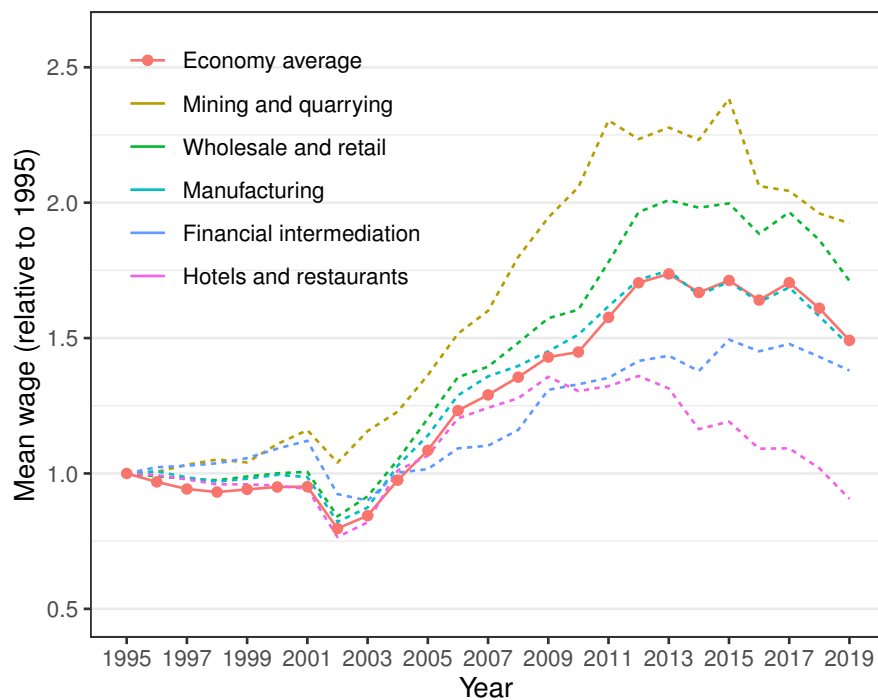


Panel C: Informality rate



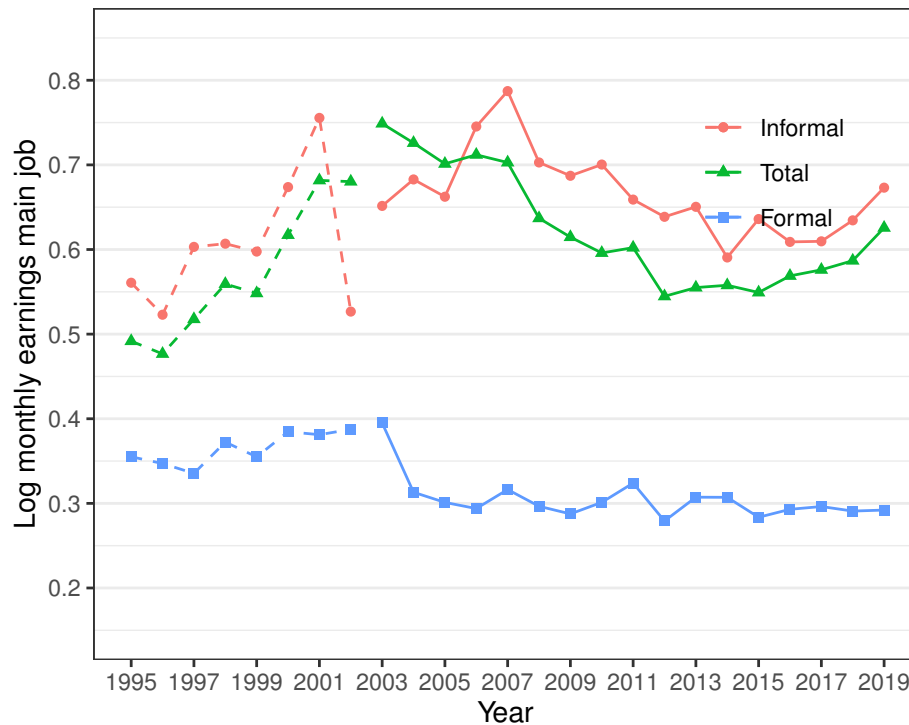
Notes: Data are from the Permanent Household Survey (EPH) of the National Institute of Statistics and Censuses (INDEC). The figure shows the evolution of key labor market indicators for Argentina between 1995 and 2019. Panel A shows the employment to population ratio, Panel B the unemployment rate (as a percentage of the economically active population), and Panel C the informality rate among salaried workers (as a percentage of the total salaried workers). The informality rate is defined as the proportion of salaried workers who report that no contributions to social security are made by their employers. The format of the EPH changed in 2003, which is indicated by different shape markers in the graphs.

Appendix Figure 2: Mean real wages across selected sectors in Argentina, 1995–2019



Notes: Data on average monthly wages by sector are obtained from the Secretary of Labor and then deflated by the CPI to obtain real values. The figure shows the evolution of average real monthly wages across selected broad sectors, each normalized to its own 1995 level. The economy average is shown as a solid line with markers.

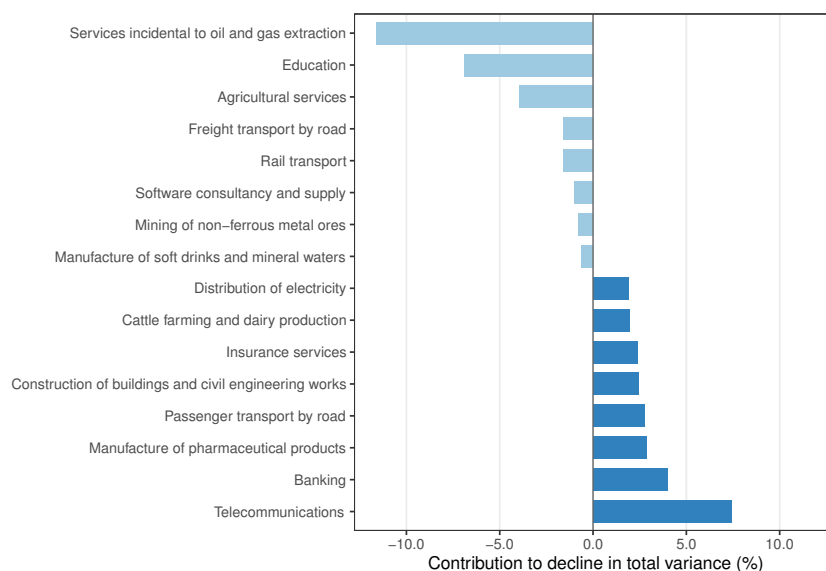
Appendix Figure 3: Variance of log earnings between 1995 and 2019, household survey data



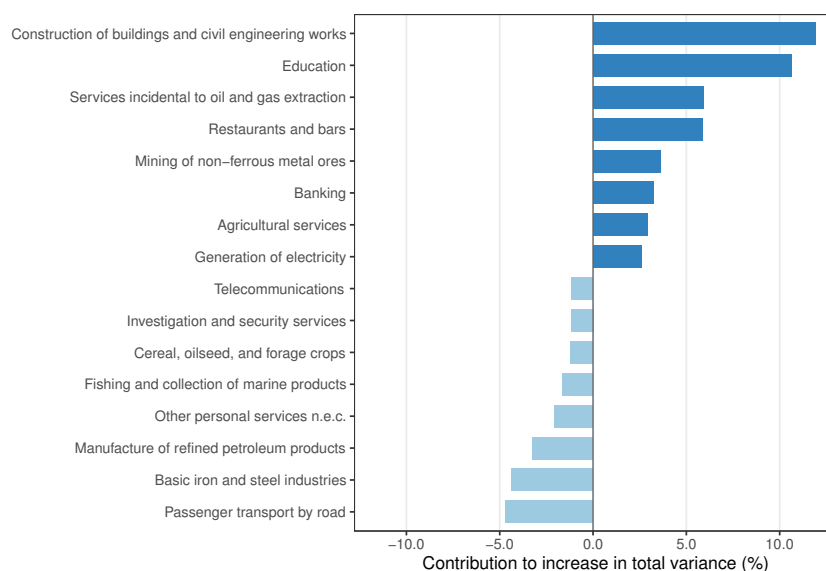
Notes: Data are from the Permanent Household Survey (EPH) for individuals aged 20 to 55. The figure shows the evolution of the variance of log real earnings in Argentina between 1995 and 2019, separately for formal, informal, and total workers. Data include earnings from the primary job of the worker only.

Appendix Figure 4: Industry contributions to changes in between-industry earnings inequality, prime-aged men

Panel A: 1995–2009



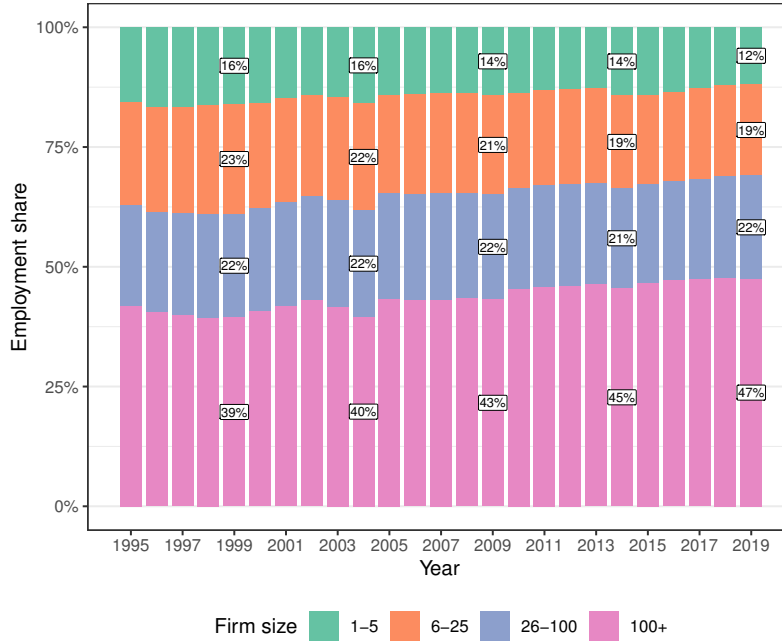
Panel B: 2005–2019



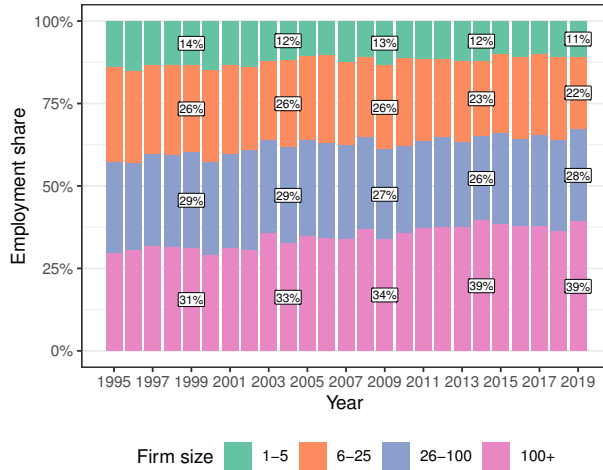
Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure shows the eight industries with the largest and the eight industries with the smallest level contributions to the change in the between-industry component of the variance of log earnings. Panel A compares 1995–1996 with 2008–2009, and Panel B compares 2005–2006 with 2018–2019. Industry employment shares and mean log earnings are calculated by pooling observations over the two years at each endpoint. An industry's contribution is the change between endpoints in the product of its employment share and its squared mean log earnings differential relative to the overall mean. Industries are selected and ordered using this contribution in levels. The values displayed divide each level contribution by the change in the total variance of log earnings. The sample excludes earnings above the 99.9th percentile or below 75 USD per month, jobs lasting fewer than six months, and internships.

Appendix Figure 5: Employment share by firm size bin

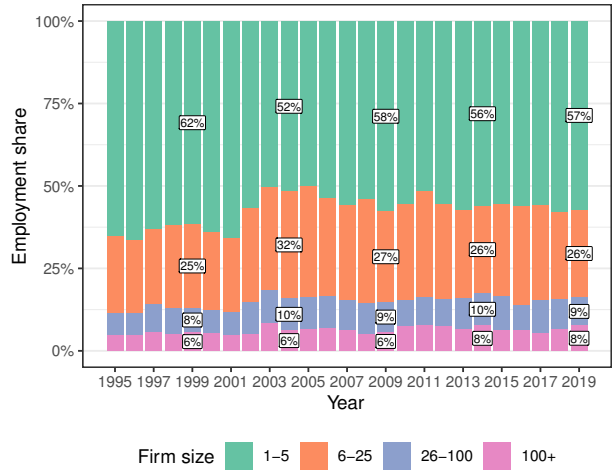
Panel A: Administrative data SIPA



Panel B: Formal workers in EPH



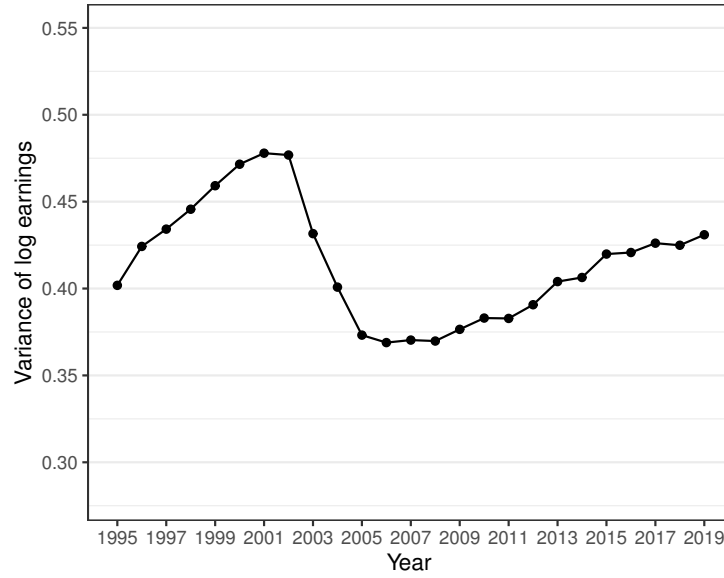
Panel C: Informal workers in EPH



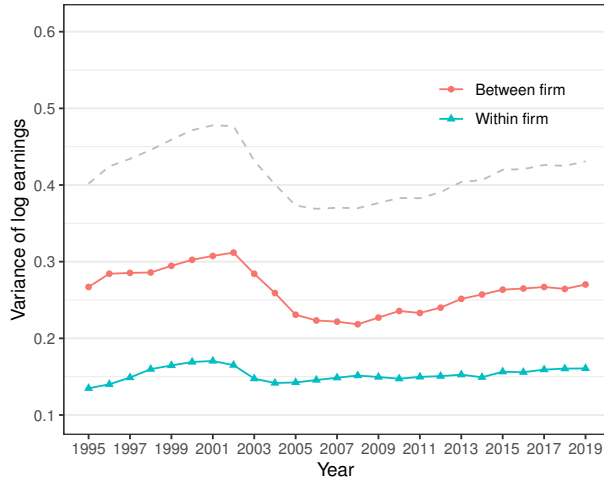
Notes: Data are from Argentina's social security system (SIPA) and from the Permanent Household Survey (EPH), for individuals aged 20 to 55. This figure shows the distribution of the number of workers across firm size bins using stacked bars, explicitly indicating the percentages for the years 1995, 2004, 2009, 2014, and 2019. In Panel A, firm size bins are defined by the number of employees at the firm: 1–5, 6–25, 26–100, and more than 100. In Panels B and C, workplace size bins are defined by the number of employees at the workplace, as declared by the worker.

Appendix Figure 6: Variance of log earnings in the formal labor market, prime-aged women

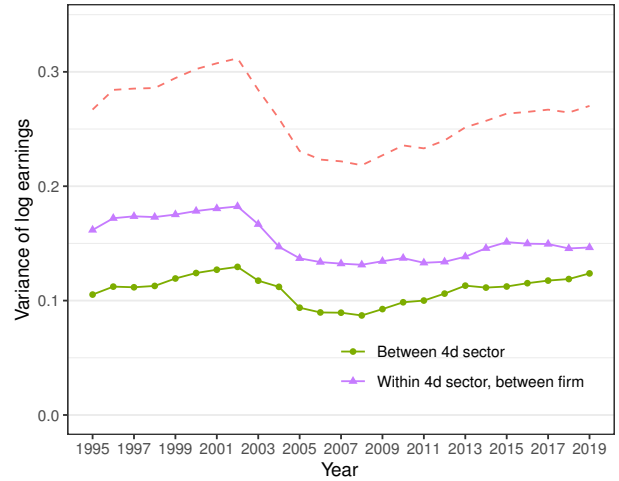
Panel A: Overall variance



Panel B: Between- and within-firm components



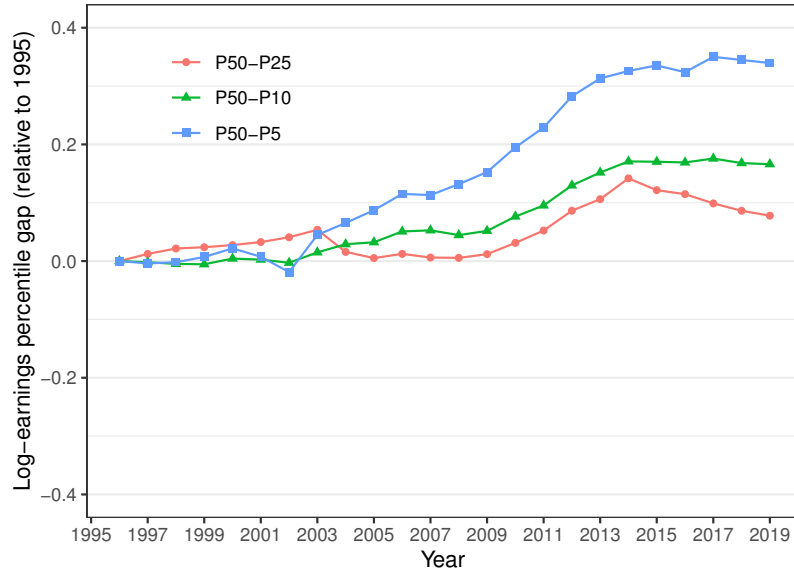
Panel C: Between-industry, within-industry between-firm, and within-firm components



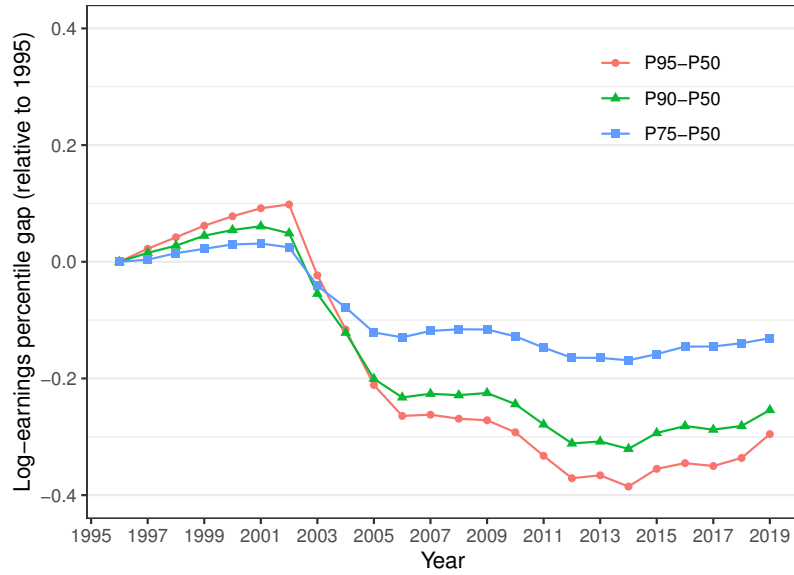
Notes: Data are from Argentina's social security system (SIPA) for women aged 20 to 55. The figure shows the evolution of inequality, defined as the variance of log earnings, between 1995 and 2019. Panel (a) shows the overall variance of log earnings, while Panels (b) and (c) show the decomposition of the variance into between- and within-firm components, and between-industry, within-industry between-firm, and within-firm components, respectively. Earnings are measured as average monthly total real earnings. Data exclude earnings above the 99.9th percentile of the distribution, and below 75 USD per month, and worker-firm-year cells that correspond to jobs shorter than 6 months and those labeled as internships.

Appendix Figure 7: Percentile gaps in log earnings in Argentina, prime-aged women

Panel A: Lower-tail inequality

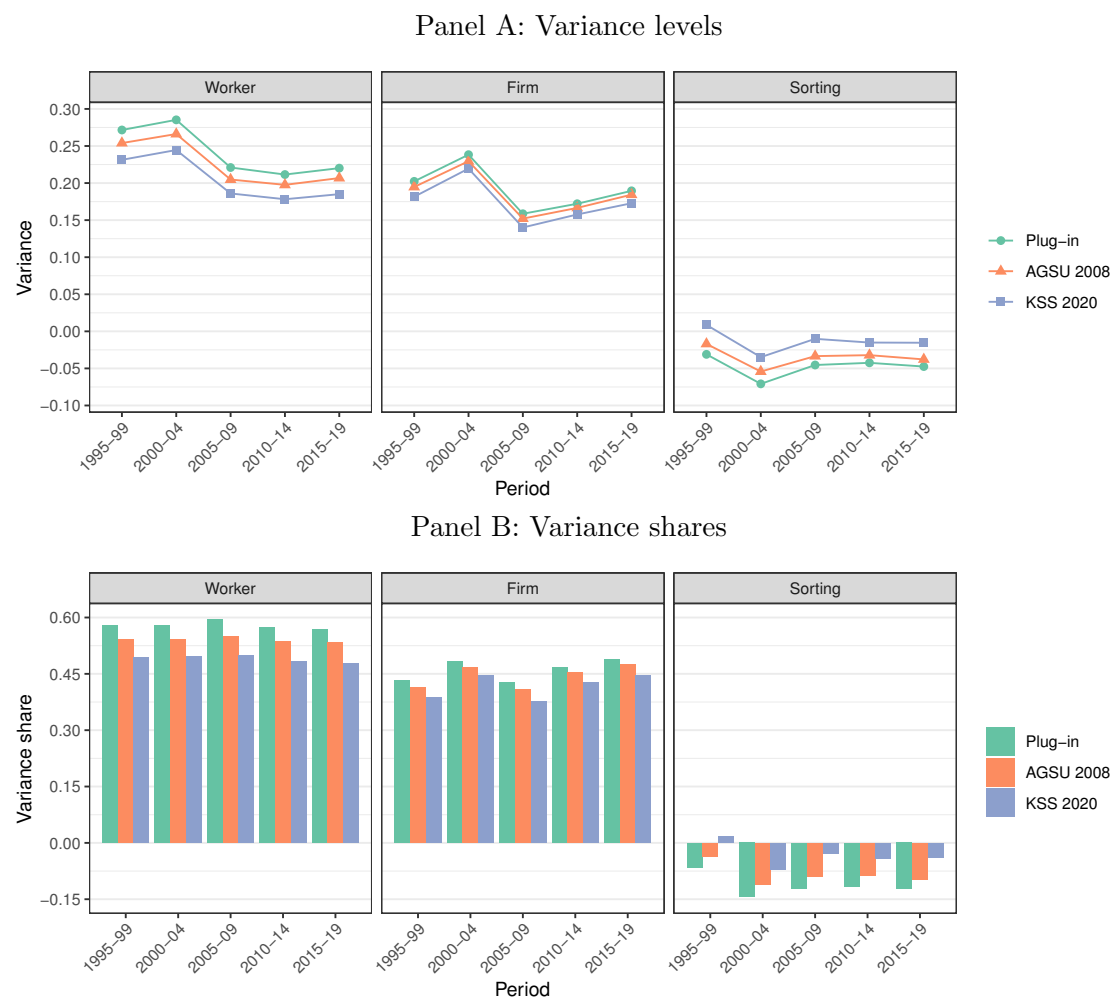


Panel B: Upper-tail inequality



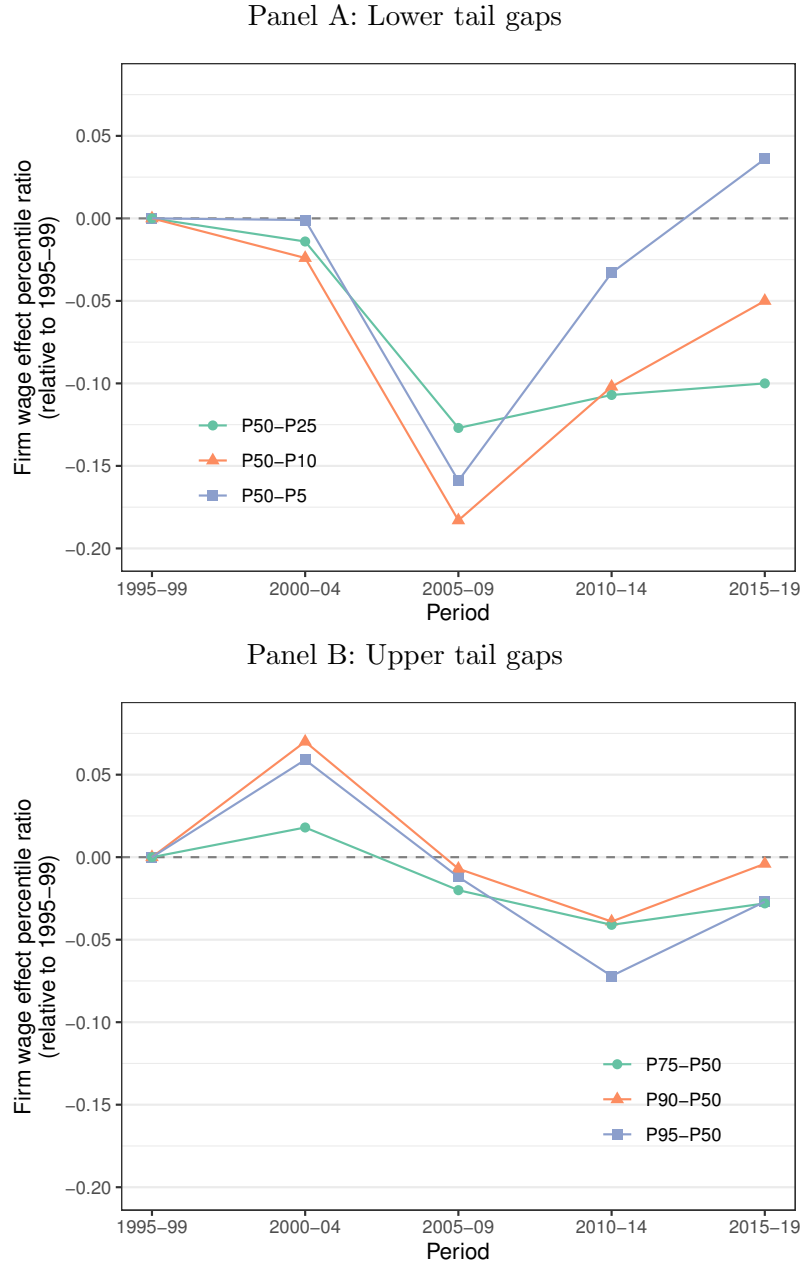
Notes: Data are from Argentina's social security system (SIPA) for women aged 20 to 55. The figure shows the evolution of inequality as measured by differences between percentiles of the log real earnings distribution. Earnings are measured as average monthly total real earnings. We exclude earnings above the 99.9th percentile of the distribution, and below 75 USD per month, and worker-firm-year cells that correspond to jobs lasting fewer than six months or internships.

Appendix Figure 8: Corrected variance decomposition of AKM estimates on leave-out-match connected set, prime-aged women



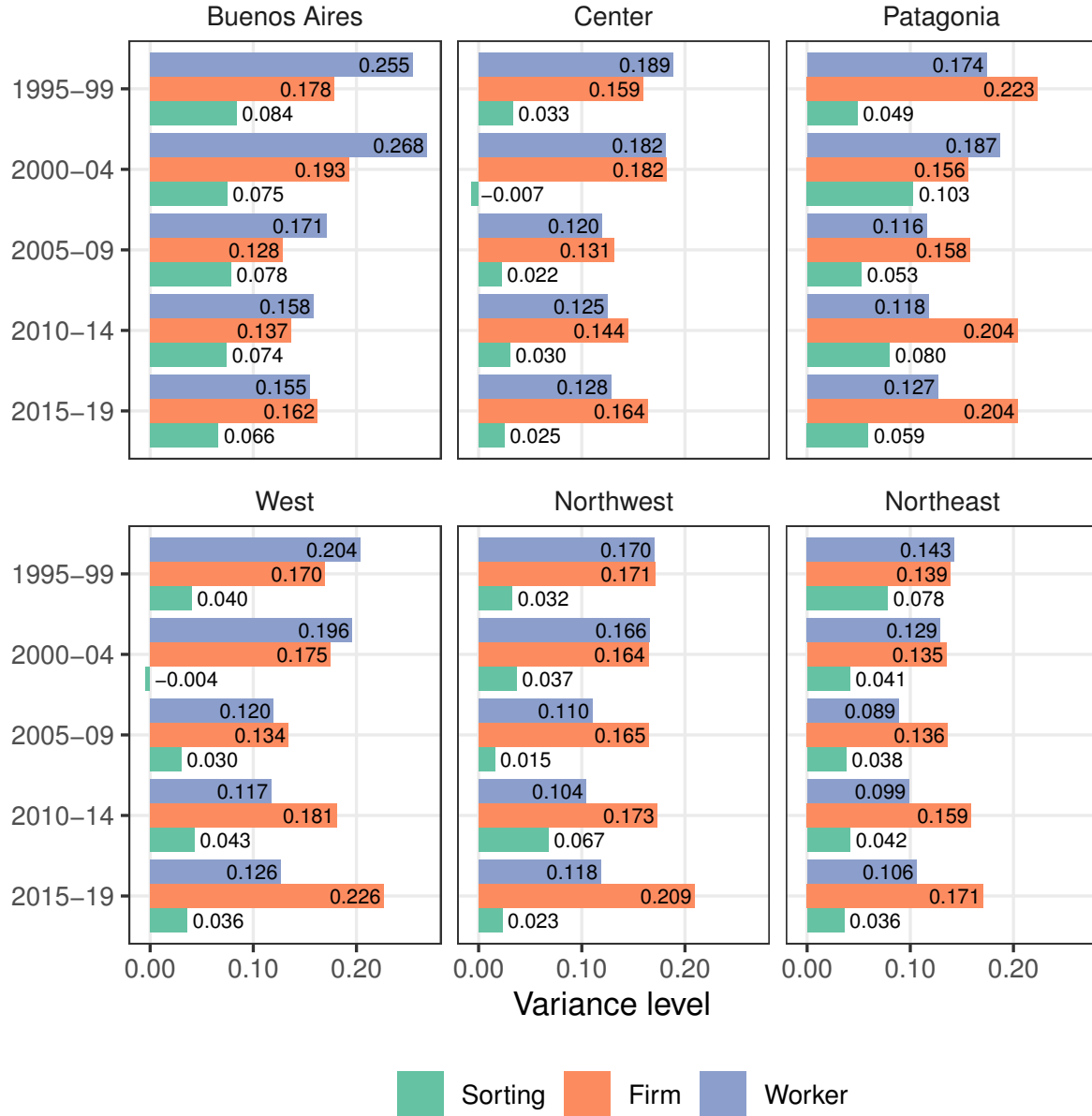
Notes: Data are from Argentina's social security system (SIPA) for women aged 20 to 55. The figure shows the contribution of each component of the variance decomposition of log earnings, correcting the estimates of the variance terms, on the leave-out-match largest connected set of firms. "Plug-in" refers to the computation of the variance components directly using the AKM estimates. "AGSU 2008" refers to the variance decomposition proposed by [Andrews et al. \(2008\)](#), which assumes that the error term is homoskedastic. "KSS 2020" refers to the variance decomposition proposed by [Kline et al. \(2020\)](#), which allows for heteroskedasticity in the error term.

Appendix Figure 9: Percentile gaps of the firm wage effect distribution



Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure shows the evolution of percentile gaps of the estimated firm wage effect distribution across the five sub-periods, normalized relative to 1995–1999 (so that all gaps equal zero in the first period). Panel A shows lower tail gaps: P50–P5, P50–P10, and P50–P25. Panel B shows upper tail gaps: P95–P50, P90–P50, and P75–P50.

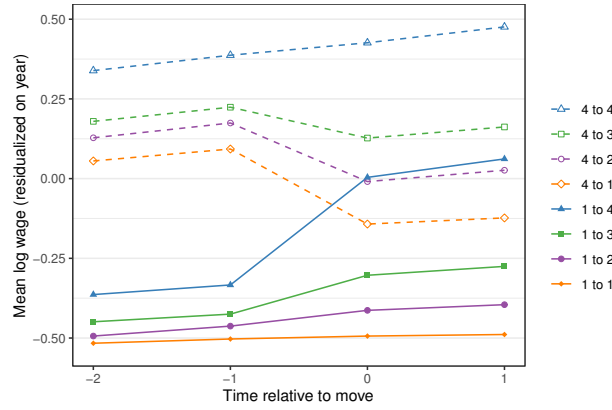
Appendix Figure 10: AKM-based variance decomposition by region



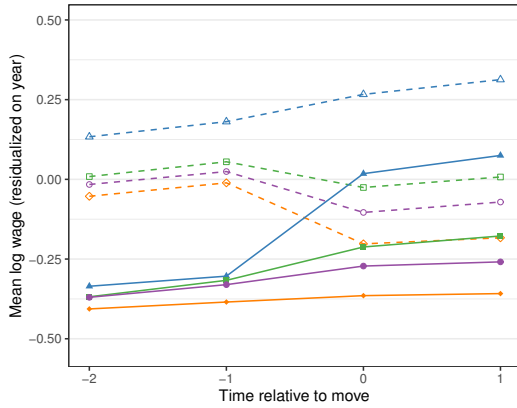
Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure shows the AKM variance decomposition by geographical region. The regions are defined as follows: "Buenos Aires" includes both the City of Buenos Aires and the Province of Buenos Aires; "Center" includes the provinces of Entre Ríos, Santa Fe, Córdoba, and La Pampa; "Patagonia" includes Río Negro, Neuquén, Chubut, Santa Cruz and Tierra del Fuego; "West" includes Mendoza, San Luis, La Rioja, and San Juan; "Northwest" includes Jujuy, Salta, Catamarca, Tucumán, and Santiago del Estero; and "Northeast" includes Misiones, Corrientes, Chaco, and Formosa.

Appendix Figure 11: Evolution of wages for workers moving across firms, prime-aged men

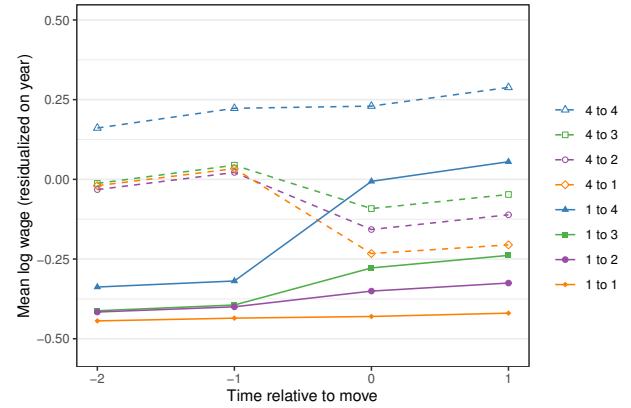
Panel A: 1997–2001



Panel B: 2006–2010

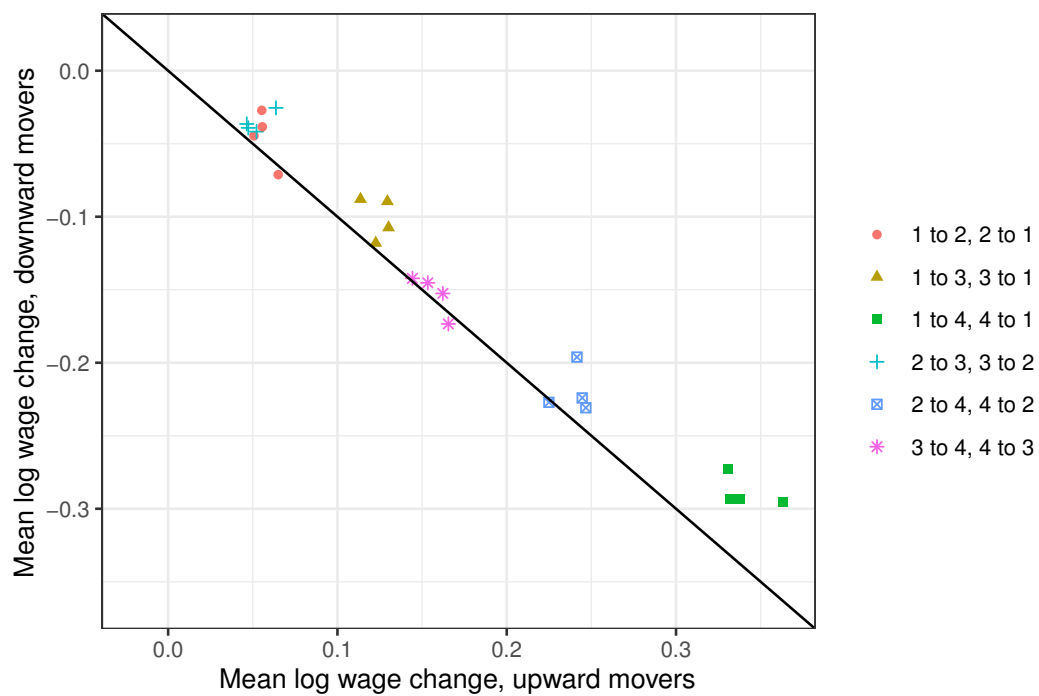


Panel C: 2013–2017



Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure shows the evolution of mean log wages (residualized on year dummies) for workers who move across firms around the move, by quartiles of co-workers' wages in the origin and destination firms. Only movers that are observed during 4 consecutive years around the move are included.

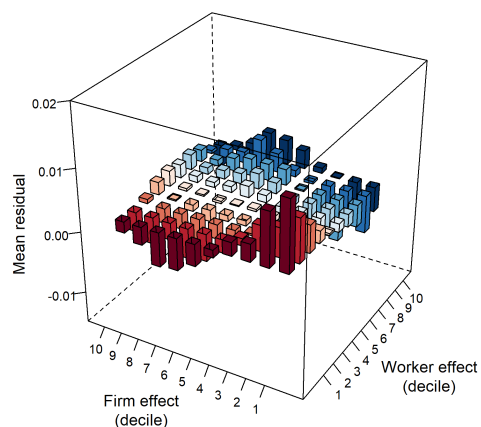
Appendix Figure 12: Evolution of wages for workers moving across firms, prime-aged men



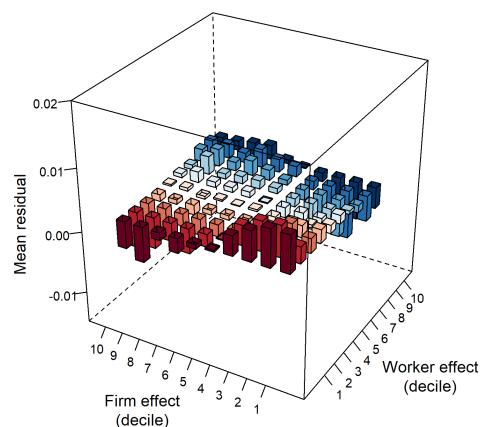
Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. Each point indicates the mean log wage change of workers who move from a firm located in a given wage quartile to another firm located in another wage quartile and for those who move in the opposite direction. Firms are classified in quartiles based on the average wage of co-workers. The black line, representing an inverted 45-degree line, is shown for reference.

Appendix Figure 13: AKM residuals across deciles of worker and firm fixed effects, prime-aged men

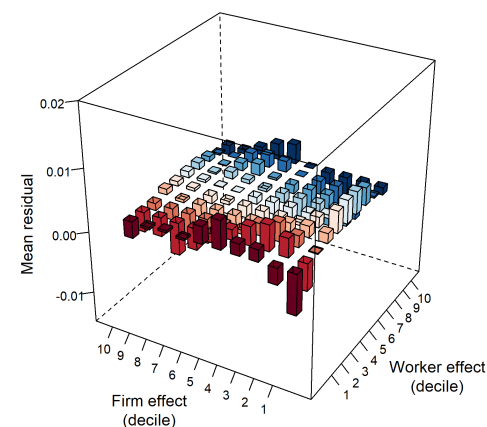
Panel A: 1995–1999



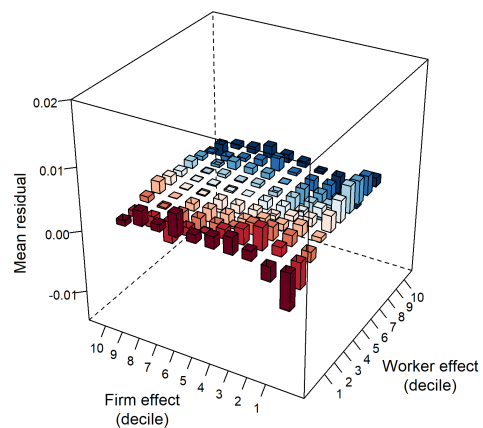
Panel B: 2000–2004



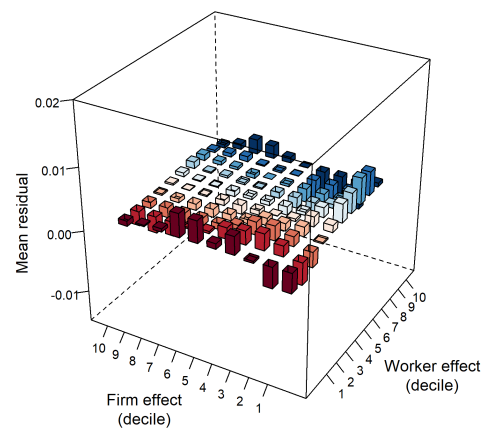
Panel C: 2005–2009



Panel D: 2010–2014

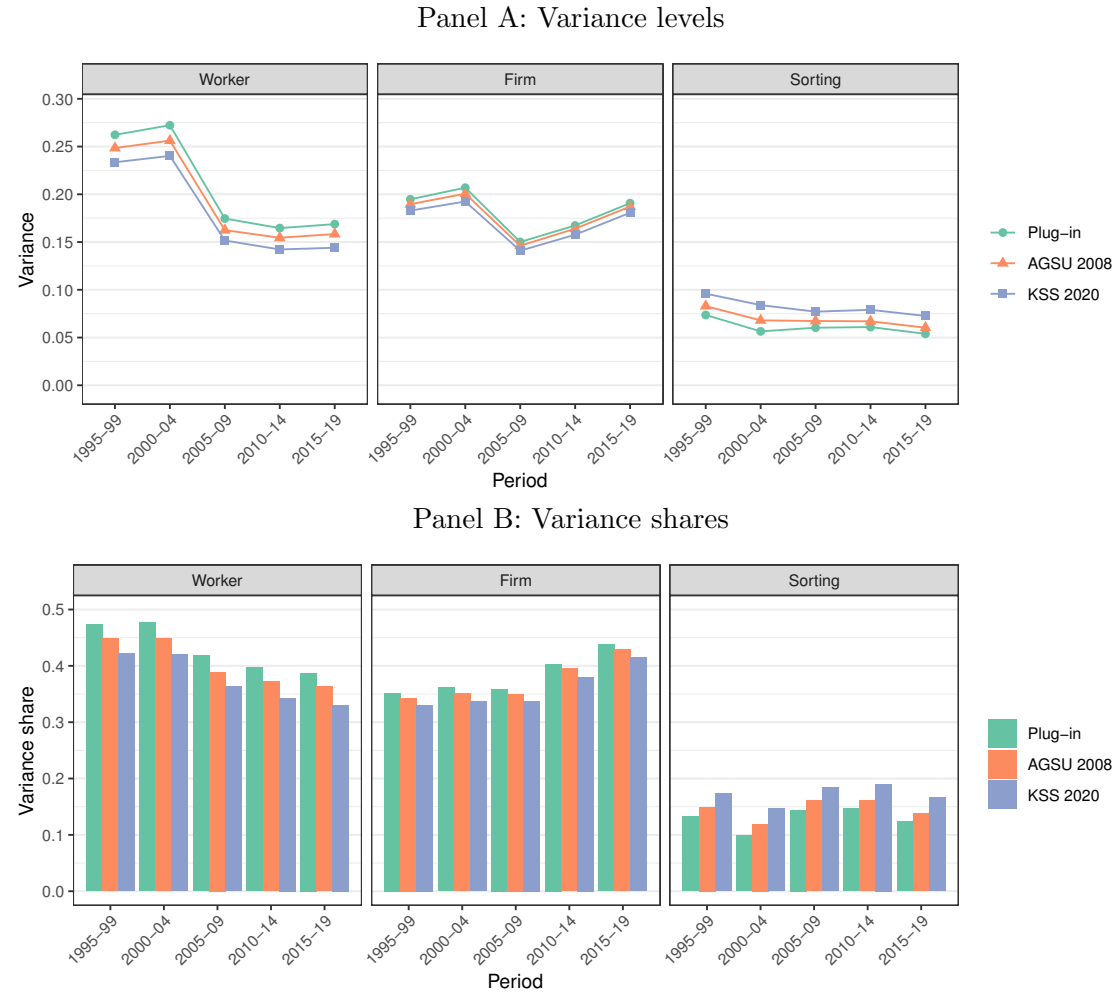


Panel E: 2015–2019



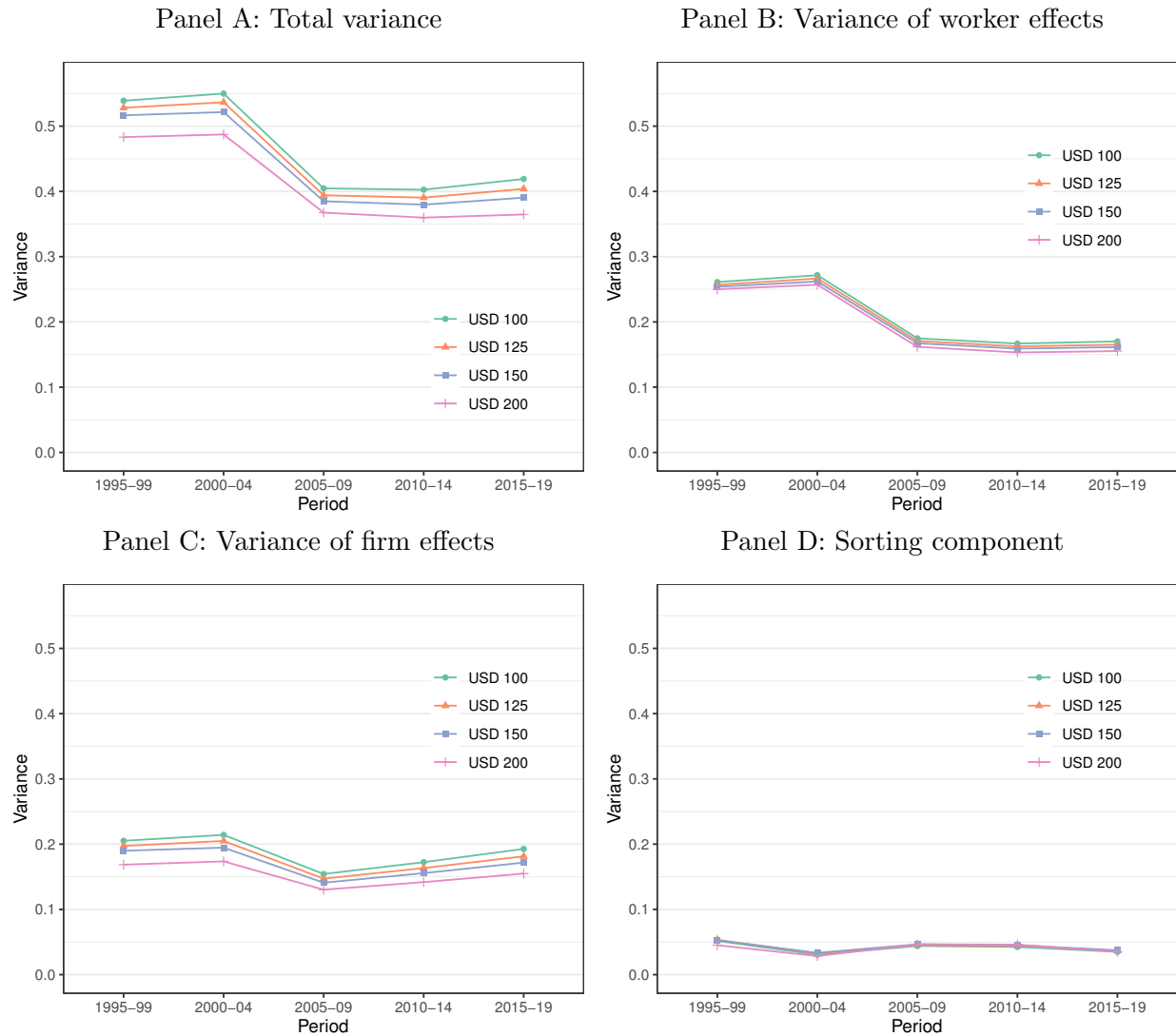
Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure shows the mean residual of the estimated AKM models by deciles of worker and firm effects. Panels A, B, C, D, and E show, respectively, mean residuals of models estimated in 1995–1999, 2000–2004, 2005–2009, 2010–2014, and 2015–2019, using the largest connected set of firms in each period.

Appendix Figure 14: Corrected variance decomposition of AKM estimates on leave-one-out connected set, prime-aged men



Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure shows the contribution of each component of the variance decomposition of log earnings, correcting the estimates of the variance terms, on the leave-out-match largest connected set of firms. "Plug-in" refers to the computation of the variance components directly using the AKM estimates. "AGSU 2008" refers to the variance decomposition proposed by [Andrews et al. \(2008\)](#), which assumes that the error term is homoskedastic. "KSS 2020" refers to the variance decomposition proposed by [Kline et al. \(2020\)](#), which allows for heteroskedasticity in the error term.

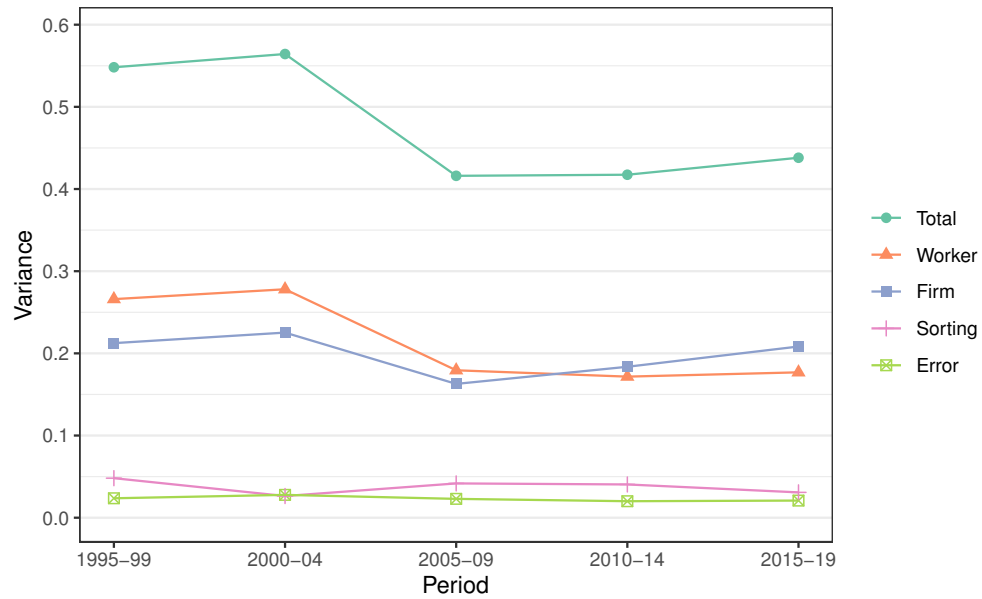
Appendix Figure 15: Variance decomposition of log earnings according to AKM model by minimum earnings cutoff, prime-aged men



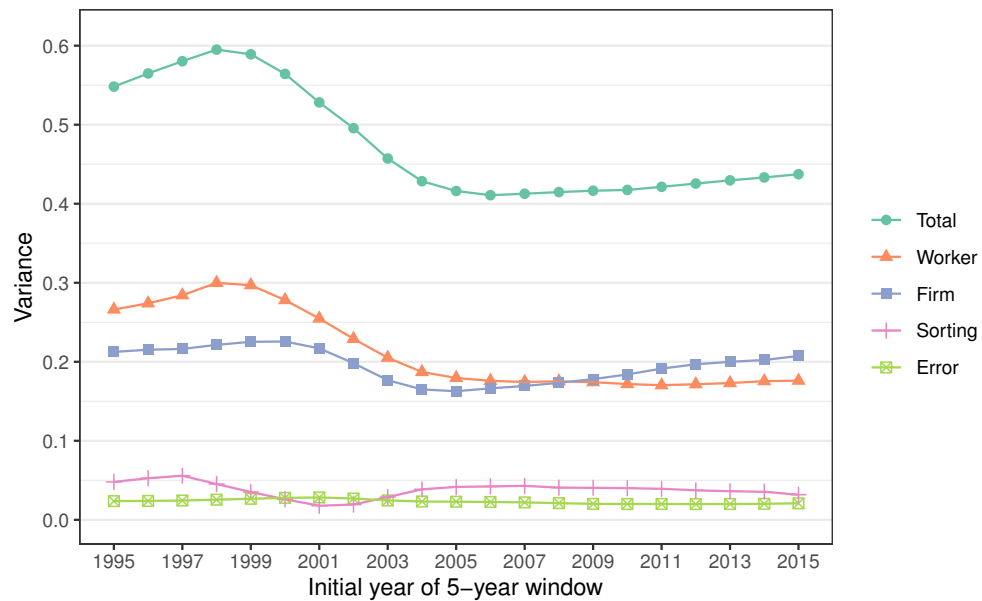
Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure shows estimates of the variance decomposition of a model of log earnings on worker fixed effects, firm fixed effects, and year fixed effects following [Abowd et al. \(1999\)](#), for different values of the level of minimum earnings to be included in the sample. The cutoff is expressed in 2010 US dollars. Panel A shows the total variance of log earnings. Panel B shows the variance of worker fixed effects. Panel C shows the variance of firm fixed effects. Panel D shows the sorting component, twice the covariance between worker and firm effects. Plug-in variance estimates are calculated on the largest connected set of firms.

Appendix Figure 16: Variance decomposition of log earnings with a time-varying window, prime-aged men

Panel A: Baseline estimates

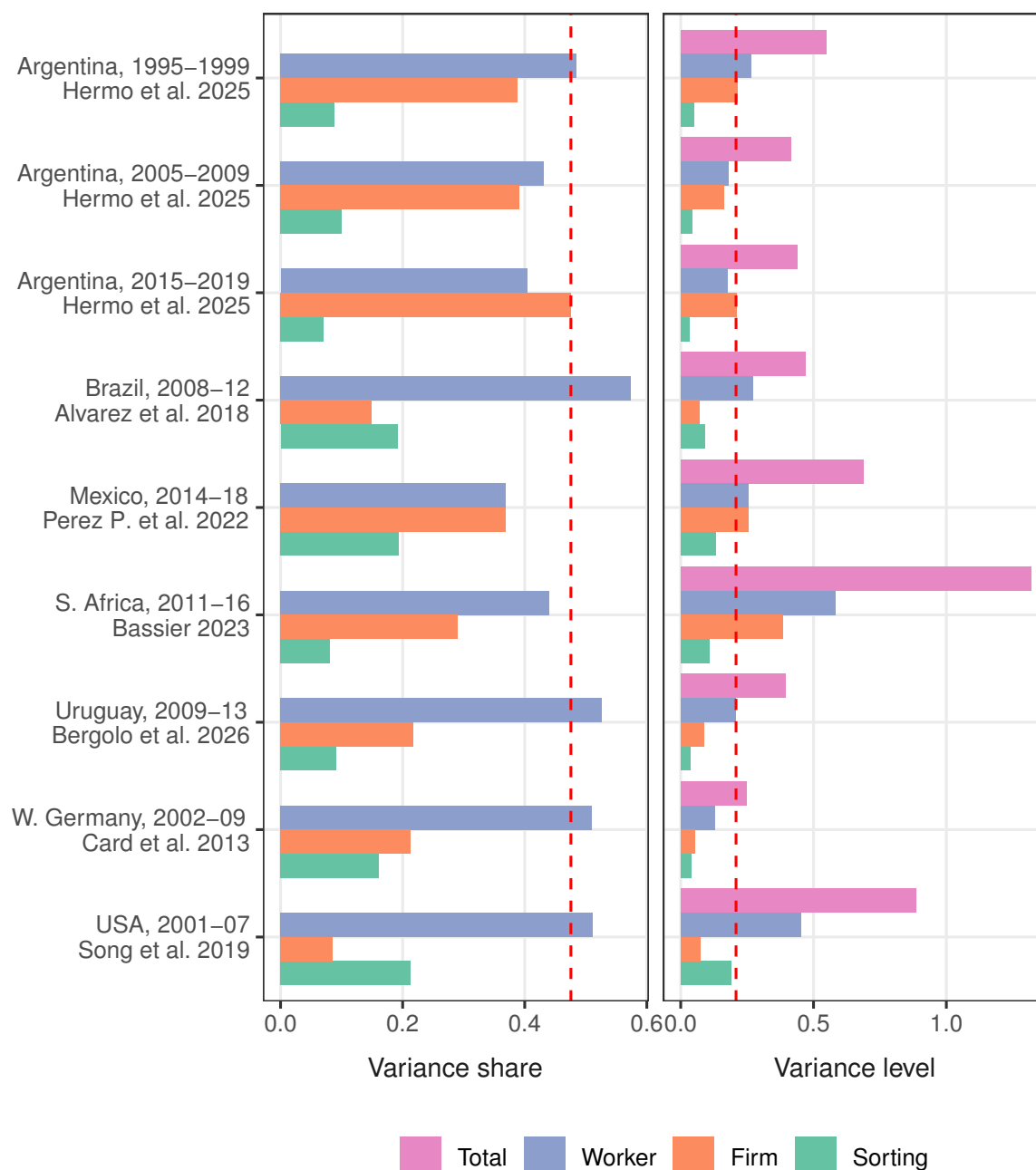


Panel B: Varying-window estimates



Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure shows estimates of the variance decomposition of a model of log earnings on worker fixed effects, firm fixed effects, and year fixed effects following [Abowd et al. \(1999\)](#), on the largest connected set of firms. Panel A replicates the baseline estimates from Table 2. Panel B shows results of models estimated separately for five-year periods with initial year ranging from 1995 through 2014.

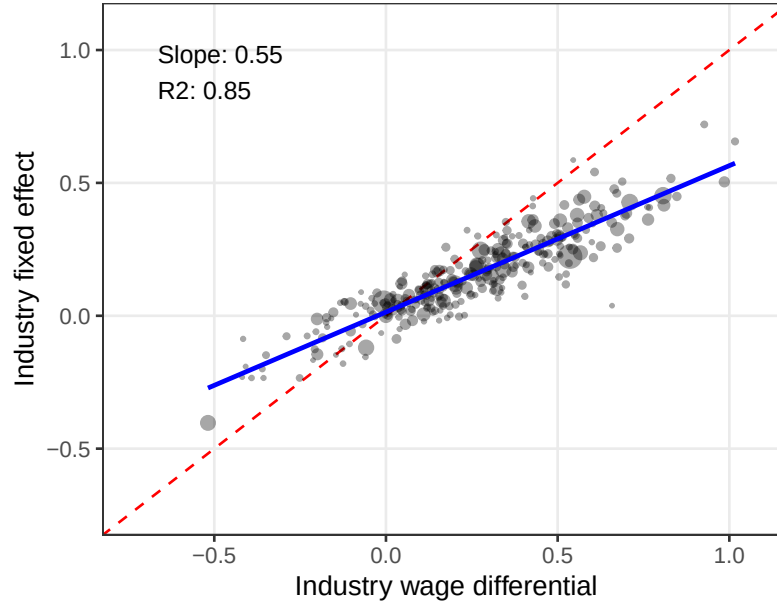
Appendix Figure 17: Comparing AKM-based variance decomposition across countries



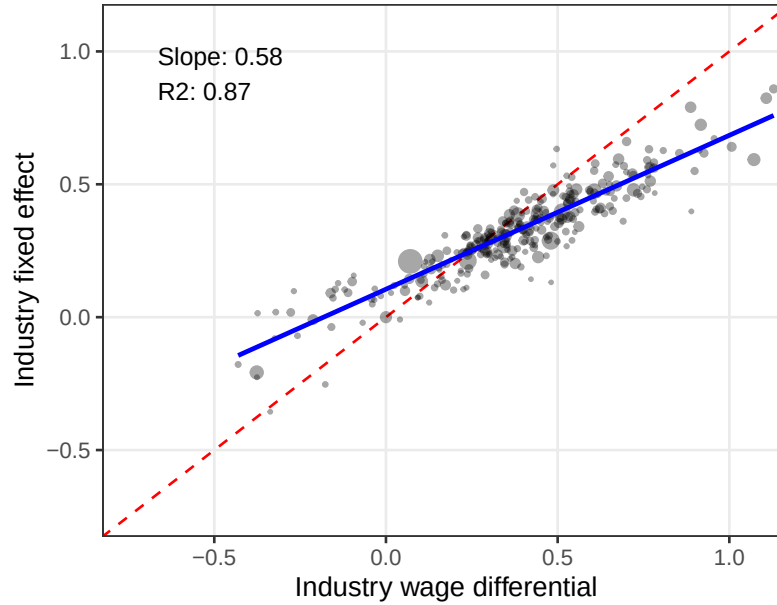
Notes: Data are from Argentina’s social security system (SIPA) for Argentina and from the selected studies cited below each country. The figure compares the AKM variance decomposition of log earnings across selected countries. The left panel shows the share of the worker, firm, and sorting components out of the total variance. The right panel shows the levels of the variances, including the total variance of log wages. The red line signals the variance level or variance share for Argentina in 2015–2019. Plug-in estimates of the variance components are chosen in all countries to ensure comparability. All reported estimations include year fixed effects as controls. [Alvarez et al. \(2018\)](#) also control for worker age, and [Card et al. \(2013\)](#) additionally control for education. All studies focus on men, with the exception of [Bassier \(2023\)](#) who includes both men and women.

Appendix Figure 18: Industry wage differentials and industry fixed effects

Panel A: 1995–99

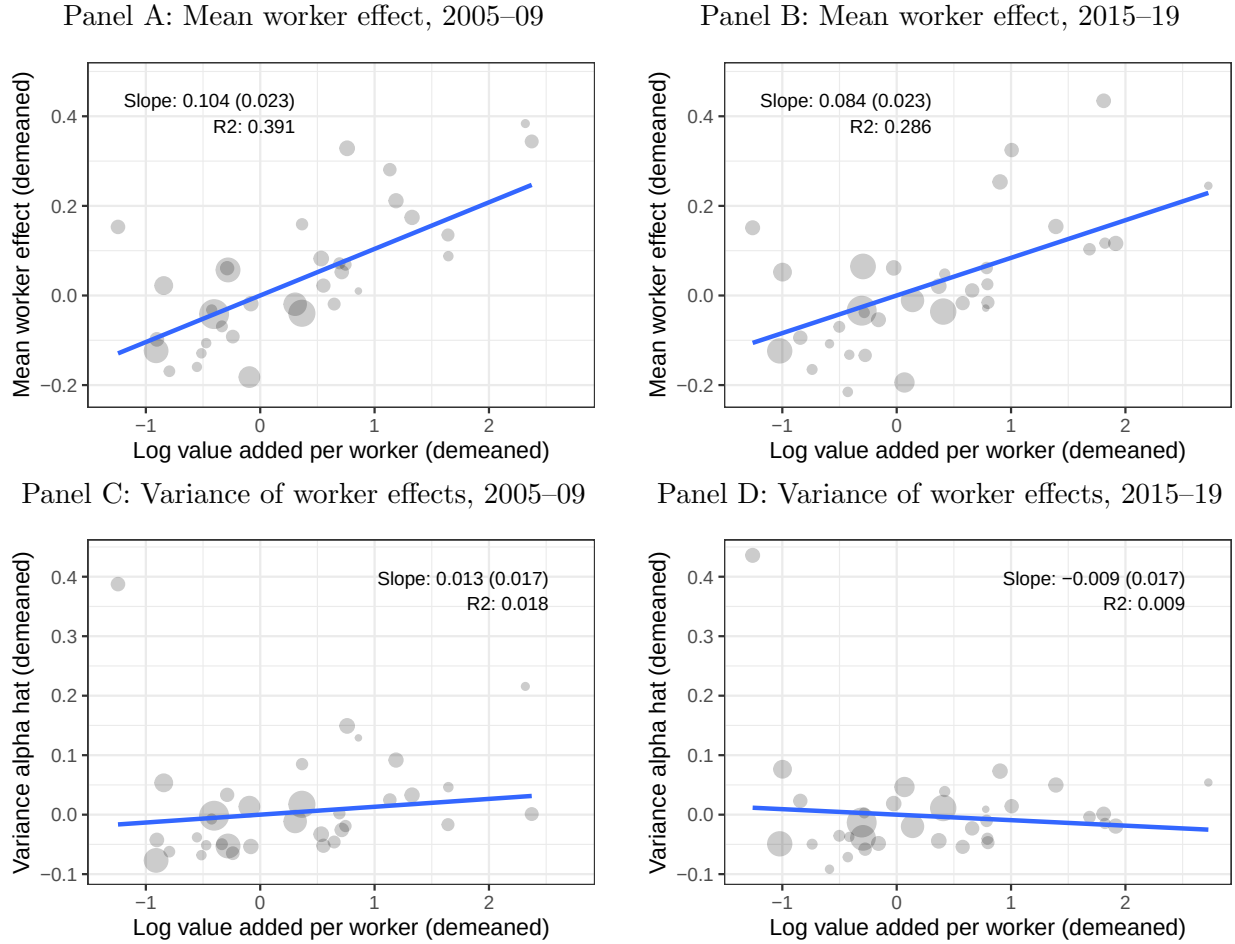


Panel B: 2015–19



Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure compares industry wage differentials with industry fixed effects at the 4-digit industry level. Industry wage differentials are computed as weighted averages of AKM firm effects within industries, normalized relative to restaurants. Industry fixed effects are obtained from a leave-out-match AKM model in which industries are treated as workplaces, again normalized relative to restaurants. Each point is a 4-digit industry and point size is proportional to worker-years. The dashed red line is the 45-degree line, while the solid blue line is a weighted linear fit.

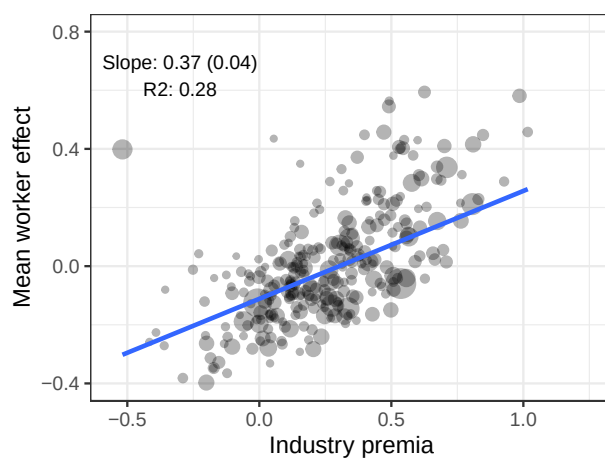
Appendix Figure 19: Worker effects and sectoral productivity in 2005–09 and 2015–19



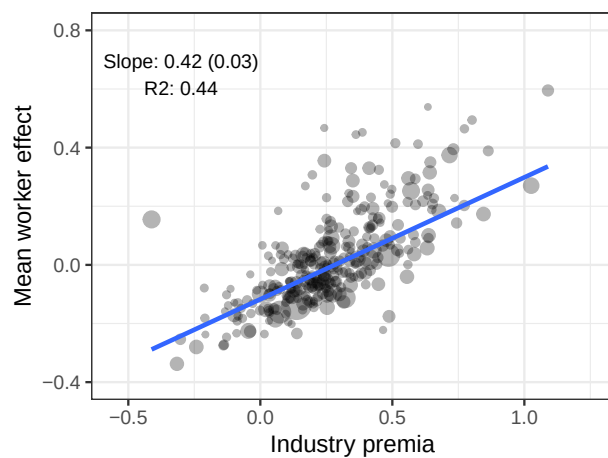
Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55 and from Centro de Estudios para la Producción (CEP XXI). The figure shows the relationship between worker effects and sectoral productivity in Argentina for the periods 2005–09 and 2015–19. Panels A and B show a scatter of the average worker effect and sectoral productivity, while Panels C and D show a scatter plot of the variance of worker effects and sectoral productivity. The measure of sectoral productivity is an average of value added per worker across relevant years. Sectors follow CEP XXI's classification based on ISIC Rev. 3.1, with selected aggregations and subdivisions.

Appendix Figure 20: Correlation between industry wage premia and worker effects

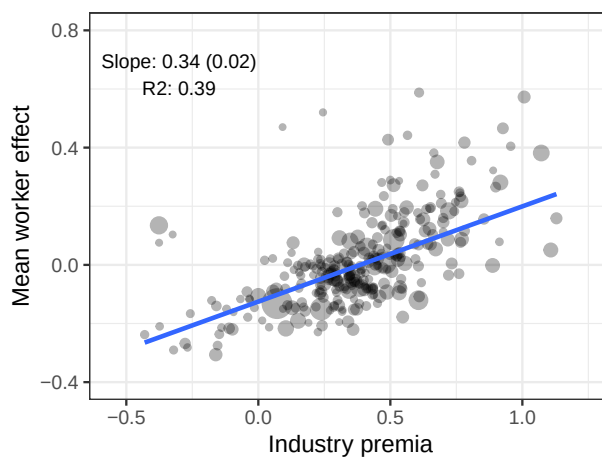
Panel A: 1995–99



Panel B: 2005–09



Panel C: 2015–19



Notes: Data are from Argentina's social security system (SIPA) for men aged 20 to 55. The figure shows the correlation between industry wage premia and the average worker effect in each industry.